Linear Algebra

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CONTENTS		
Chapter no.	**CHAPTER NAME**	**Page no.**
		4
1.	**Linear Algebra**	
	1.1 Introduction	
	1.2 Applications of Linear Algebra	
	1.3 Matrix	
	1.3.1 Vector	
	1.3.2 Submatrix	
	1.3.3 Types of matrices	
	1.3.3.1 Square matrix	
	1.3.3.2 Upper triangular matrix	
	1.3.3.3 Lower triangular matrix	
	1.3.3.4 Diagonal matrix	
	1.3.3.5 Identity matrix	
	1.3.3.6 Zero matrix	
	1.3.3.7 Tridiagonal matrix	
	1.3.3.8 Symmetric matrix	
	1.3.3.9 Skew-symmetric matrix	
	1.3.4 Addition of matrices	
	1.3.5 Subtraction of matrices	
	1.3.6 Multiplication of matrices	
	1.3.7 Linear combination of matrices	
	1.3.8 Rule of binary matrix operations	
	1.3.8.1 Commutative law of addition	
	1.3.8.2 Associative law of addition	
	1.3.8.3 Commutative law of multiplication	
	1.3.8.4 Distributive law of multiplication	
	1.3.9 Transpose of a matrix	
	1.3.10 Trace of a matrix	
	1.4 Determinant of a matrix	
	1.4.1 Calculation of determinant of any square matrix	
	1.4.2 Theorems related to determinant of square matrix	
	1.4.3 Properties of determinant of square matrix	
	1.5 System of Linear equations	
	1.6 Rank of a matrix	
	1.7 Consistent and inconsistent system of Linear equations	
	1.8 Inverse of a matrix	
	1.9 Eigen values and Eigen vectors	
	1.9.1 Methods for finding Eigen values of a square matrix	
	1.9.2 Theorems related to Eigen values and Eigen vectors	
	1.10 Caley-Hamilton theorem	

Career Avenues

	SET-1	
	SET-2	
	SET-3	
	Past GATE Questions	

Career Avenues

Chapter-1 Linear Algebra

1.1 Introduction

Linear algebra is a branch of mathematics that deals with mathematics of data and the language of data are Matrices and vectors. It is all about linear combinations, by using arithmetic on columns of numbers called vectors and arrays of numbers called matrices, to create new columns and arrays of numbers.

Linear algebra is the study of lines and planes, vector spaces and mappings that are required for linear transforms.

It is a relatively young field of study, having initially been formalized in the 1800s in order to find unknowns in systems of linear equations. A linear equation is just a series of terms and mathematical operations

1.2 Applications of Linear Algebra

As linear algebra is the mathematics of data, the tools of linear algebra are used in many domains. Briefly they are:

- Matrices in Engineering, such as a line of springs.
- Graphs and Networks, such as analysing networks.
- Markov Matrices, Population, and Economics, such as population growth.
- Linear Programming, the simplex optimization method.
- Fourier Series: Linear Algebra for functions, used widely in signal processing.
- Linear Algebra for statistics and probability, such as least squares for regression.
- Computer Graphics, such as the various translation, rescaling and rotation of images.

1.3 Matrix:-

- A matrix is a rectangular array of elements.
- It is a collection of numbers ordered by rows and columns.
- It is customary to enclose the elements of a matrix in parentheses, brackets, or braces. The elements can be symbolic expressions or numbers.
- Matrix ***[A]*** is denoted by

$$[A]=\begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22} & ..... & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & ..... & a_{mn} \end{bmatrix}$$

Row ***i*** of ***[A]*** has ***n*** elements and is

$$\begin{bmatrix} a_{i1} & a_{i2}....a_{in} \end{bmatrix}$$

and column ***j*** of ***[A]*** has ***m*** elements and is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

- Each matrix has rows and columns and this defines the size of the matrix.

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- If a matrix ***[A]*** has ***m*** rows and n columns, the size of the matrix is denoted by ***m x n***. The matrix ***[A]*** may also be denoted by $[A]_{m \times n}$ to show that ***[A]*** is a matrix with ***m*** rows and ***n*** columns.

(That is, the first subscript in a matrix refers to the row and the second subscript refers to the column. It is important to remember this convention when matrix algebra is performed.

- Each entry in the matrix is called the entry or element of the matrix and is denoted by a_{ij} where ***i*** is the row number and ***j*** is the column number of the element.

For example: The matrix for the jeans sales example could be denoted by the matrix [A] as

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}.$$

There are 3 rows and 4 columns, so the size of the matrix is 3 x 4. In the above *[A]* matrix, a_{34} = 27.

1.3.1 Vector:

- A vector is a special type of matrix that has only one row (called a row vector) or one column (called a column vector).
- Matrix **[B]** has one row, it is called a row vector

$$[B] = [b_1 \; b_2 \ldots\ldots b_n]$$

and ***n*** is the dimension of the row vector.

Example of a row vector of dimension 5: [B] = [25 20 3 2 0]

- Matrix ***[C]*** has one column, it is called a column vector

$$[C] = \begin{bmatrix} c_1 \\ \vdots \\ \vdots \\ c_m \end{bmatrix}$$

Where ***m*** is the dimension of the vector.

- ***Example of a column vector of dimension 3:***

$$[C] = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

Career Avenues

1.3.2 Submatrix

If some row(s) or/and column(s) of a matrix ***[A]*** are deleted, the remaining matrix is called a submatrix of ***[A]***.

For matrix,

$$[A]=\begin{bmatrix}4 & 6 & 2\\ 3 & -1 & 2\end{bmatrix}$$

Submatrices of ***[A]***

$$\begin{bmatrix}4 & 6 & 2\\ 3 & -1 & 2\end{bmatrix},\begin{bmatrix}4 & 6\\ 3 & -1\end{bmatrix},\begin{bmatrix}4 & 6 & 2\end{bmatrix},\begin{bmatrix}4\end{bmatrix},\begin{bmatrix}2\\ 2\end{bmatrix}$$

1.3.3 Types of Matrices

1.3.3.1 Square matrix

- If the number of rows ***m*** of a matrix is equal to the number of columns ***n*** of a matrix ***[A]***, (m =n), then ***[A]*** is called a square matrix.
- The entries $a_{11}, a_{22}, ..., a_{nn}$ are called the diagonal elements of a square matrix. Sometimes the diagonal of the matrix is also called the principal or main of the matrix.
- Example of a square matrix (same number of rows and columns, i.e., **3**)

$$[A]=\begin{bmatrix}25 & 20 & 3\\ 5 & 10 & 15\\ 6 & 15 & 7\end{bmatrix}$$

The diagonal elements of $[A]$ are $a_{11}=25,\ a_{22}=10,\ a_{33}=7$.

1.3.3.2 Upper triangular matrix

- A ***m x n*** matrix for which $a_{ij} = 0,\ i > j$ is called an upper triangular matrix.
- All the elements below the diagonal entries are zero.
- Example of an upper triangular matrix:

$$[A]=\begin{bmatrix}10 & -7 & 0\\ 0 & -0.001 & 6\\ 0 & 0 & 15005\end{bmatrix}$$

1.3.3.3 Lower triangular matrix

- A ***m x n*** matrix for which $a_{ij} = 0,\ j > i$ is called a lower triangular matrix.
- All the elements above the diagonal entries are zero.
- Example of a lower triangular matrix:

$$[A]=\begin{bmatrix}1 & 0 & 0\\ 0.3 & 1 & 0\\ 0.6 & 2.5 & 1\end{bmatrix}$$

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1.3.3.4 Diagonal matrix

- A square matrix with all non-diagonal elements equal to zero is called a diagonal matrix,
- Only the diagonal entries of the square matrix can be non-zero, ($a_{ij} = 0, i \neq j$)
- Example of a diagonal matrix:

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- ***Any or all the diagonal entries of a diagonal matrix can be zero.*** For example,

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1.3.3.5 Identity matrix

- A diagonal matrix with all diagonal elements equal to one is called an identity matrix, ($a_{ij} = 0, i \neq j$)and $\boldsymbol{a_{ii} = 1}$ for all ***i***).
- Example of an Identity matrix:

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1.3.3.6 Zero matrix

- A matrix whose all entries are zero is called a zero matrix, ($a_{ij} = 0$ for all i and j).
- Example of a zero matrix:

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1.3.3.7 Tridiagonal matrix

- A tridiagonal matrix is a square matrix in which all elements are not zero at, the major diagonal, the diagonal above the major diagonal, and the diagonal below the major diagonal.
- Example of a tridiagonal matrix:

$$[A] = \begin{bmatrix} 2 & 4 & 0 & 0 \\ 2 & 3 & 9 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

Career Avenues

1.3.3.8 Symmetric matrix

- A square matrix $[A]$ with real elements where $a_{ij} = a_{ji}$ for $i = 1, 2, ..., n$ and $j = 1, 2, ..., n$ is called a symmetric matrix. This is same as, if $[A]=[A]^T$, then $[A]^T$ is a symmetric matrix.
- Example of a symmetric matrix:

$$[A]=\begin{bmatrix} 21.2 & 3.2 & 6 \\ 3.2 & 21.5 & 8 \\ 6 & 8 & 9.3 \end{bmatrix}$$

as $a_{12} = a_{21} = 3.2, a_{13} = a_{31} = 6$ and $a_{23} = a_{32} = 8$

1.3.3.9 Skew-symmetric matrix:

- A $n \times n$ matrix is skew symmetric if $a_{ij} = -a_{ji}$ for i = 1,..., n and j = 1,..,n

  i.e., $[A]=-[A]^T$

- Example of a Skew-symmetric matrix:

$$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -5 \\ -2 & 5 & 0 \end{bmatrix}$$

As $a_{12} = -a_{21} = 1; a_{13} = -a_{31} = 2; a_{23} = -a_{32} = -5$ Since $a_{ii} = -a_{ii}$ only if $a_{ii} = 0$ all the diagonal elements of a skew-symmetric matrix have to be zero.

1.3.4 Addition of Matrices:

- Two matrices $[A]$ and $[B]$ can be added only if they are the same size. The addition is then shown as

$$[C]=[A]+[B]$$

Where $c_{ij} = a_{ij} + b_{ij}$

Example: Add the following two matrices.

$$[A]=\begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} \qquad [B]=\begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

Solution:

$$[C]=[A]+[B]$$

$$=\begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}+\begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

$$=\begin{bmatrix} 5+6 & 2+7 & 3-2 \\ 1+3 & 2+5 & 7+19 \end{bmatrix}$$

Career Avenues

$$= \begin{bmatrix} 11 & 9 & 1 \\ 4 & 7 & 26 \end{bmatrix}$$

1.3.5 Subtraction of two matrices

- Two matrices $[A]$ and $[B]$ can be subtracted only if they are the same size. The subtraction is then given by

$$[D]=[A]-[B]$$

Where $d_{ij} = a_{ij} - b_{ij}$

Example: Subtract matrix $[B]$ from matrix $[A]$.

$$[A]=\begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B]=\begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

Solution:

$$[D]=[A]-[B]$$

$$= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} - \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

$$= \begin{bmatrix} (5-6) & (2-7) & (3-(-2)) \\ (1-3) & (2-5) & (7-19) \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -5 & 5 \\ -2 & -3 & -12 \end{bmatrix}$$

1.3.6 Multiplication of two Matrices: -

- Two matrices $[A]$ and $[B]$ can be multiplied only if the number of columns of $[A]$ is equal to the number of rows of $[B]$ to give $[C]_{m\times n} = [A]_{m\times p}[B]_{p\times n}$
- If $[A]$ is a $m\times p$ matrix and $[B]$ is a $p\times n$ matrix, the resulting matrix $[C]$ is a $m\times n$ matrix.

So how does one calculate the elements of $[C]$ ***matrix?***

$$cij = \sum_{k=1}^{p} a_{ik} b_{kj}$$

To put it in simpler terms, the ith row and j^{th} column of the $[C]$ matrix in $[C]=[A][B]$ is calculated by multiplying the ith row of $[A]$ by the jth column of $[B]$, that is,

$$c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \ldots\ldots & a_{ip} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ : \\ : \\ b_{pj} \end{bmatrix}$$

$$= a_{i1} b_{1j} + a_{i2} b_{2j} + \ldots\ldots\ldots + a_{ip} b_{pj}$$

Career Avenues

$$= \sum_{k=1}^{p} a_{ik}\, b_{kj}$$

Example: Given

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 3 & -2 \\ 5 & -8 \\ 9 & -10 \end{bmatrix}$$

Find

$$[C] = [A][B]$$

Solution:

c_{12} can be found by multiplying the first row of $[A]$ by the second column of $[B]$,

$$c_{12} = \begin{bmatrix} 5 & 2 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -8 \\ -10 \end{bmatrix}$$

$$= (5)(-2) + (2)(-8) + (3)(-10)$$

$$= -56$$

Similarly, one can find the other elements of $[C]$ to give

$$[C] = \begin{bmatrix} 52 & -56 \\ 76 & -88 \end{bmatrix}$$

Scalar product of a constant and a matrix

If $[A]$ is a $n \times n$ matrix and k **is a real number, then the scalar product of** k **and $[A]$ is another $n \times n$ matrix** [B] **, where $b_{ij} = k\, a_{ij}$.**

Example:

Let

$$[A] \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}$$

Find $2[A]$

Solution:

$$2[A] = 2 \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 2.1 & 2 \times 3 & 2 \times 2 \\ 2 \times 5 & 2 \times 1 & 2 \times 6 \end{bmatrix} \qquad = \begin{bmatrix} 4.2 & 6 & 4 \\ 10 & 2 & 12 \end{bmatrix}$$

1.3.7 Linear combination of matrices:

- If $[A_1], [A_2], ..., [A_p]$ are matrices of the same size and $k_1, k_2, ..., k_p$ are scalars, then $k_1[A_1] + k_2[A_2] + ... + k_p[A_p]$ is called a linear combination of $[A_1], [A_2], ..., [A_p]$

Career Avenues

Example:

If

$$[A_1]=\begin{bmatrix}5 & 6 & 2\\ 3 & 2 & 1\end{bmatrix},[A_2]=\begin{bmatrix}2.1 & 3 & 2\\ 5 & 1 & 6\end{bmatrix},[A_3]=\begin{bmatrix}0 & 2.2 & 2\\ 3 & 3.5 & 6\end{bmatrix}$$

Then find, $[A_1]+2[A_2]=0.5[A_3]$

Solution:

$$[A_1]+2[A_2]=0.5[A_3]$$

$$=\begin{bmatrix}5 & 6 & 2\\ 3 & 2 & 1\end{bmatrix}+2\begin{bmatrix}2.1 & 3 & 2\\ 5 & 1 & 6\end{bmatrix}-0.5\begin{bmatrix}0 & 2.2 & 2\\ 3 & 3.5 & 6\end{bmatrix}$$

$$=\begin{bmatrix}5 & 6 & 2\\ 3 & 2 & 1\end{bmatrix}+\begin{bmatrix}4.2 & 6 & 4\\ 10 & 2 & 12\end{bmatrix}-\begin{bmatrix}0 & 1.1 & 1\\ 1.5 & 1.75 & 3\end{bmatrix}$$

$$=\begin{bmatrix}9.2 & 10.2 & 5\\ 11.5 & 2.25 & 10\end{bmatrix}$$

1.3.8 Rule of binary matrix operations

1.3.8.1 Commutative law of addition

If [A] and [B] are $m\times n$ matrices, then

[A] + [B] = [B] + [A]

1.3.8.2 Associative law of addition

If [A], [B] and [C] are all $m\times n$ matrices, then

[A] + ([B] + [C]) = ([A] + [B]) + [C]

1.3.8.3 Associative law of multiplication

If [A], [B] and [C] are $m\times n, n\times p \text{ and } p\times r$ size matrices, respectively, then

[A] ([B] [C]) = ([A][B]) [C]

and the resulting matrix size on both sides of the equation is $m\times r$.

1.3.8.4 Distributive law:

If [A] and [B] are $m\times n$ size matrices, and [C] and [D] are $n\times p$ size matrices

[A] ([C] + [D]) = [A] [C] + [A] [D]

([A] + [B]) [C] = [A] [C] + [B] [C]

and the resulting matrix size on both sides of the equation is $m\times p$.

1.3.9 Transpose of a matrix

- Let [A] be a $m\times n$ matrix. Then [B] is the transpose of the [A] if $b_{ij}=a_{ij}$ for all i and j.
- That is, the ith row and the jth column element of [A] is the jth row and i^{th} column element of [B]. Note, [B] would be a $n\times m$ matrix.
- The transpose of [A] is denoted by $[A]^T$.
- Properties of Transpose of Matrix

Career Avenues

(a) $\left(A^T\right)^T = A$

(b) $\left(A+B\right)^T = A^T + B^T$

(c) $\left(AB\right)^T = B^T A^T$

(d) $\left(kA\right)^T = k\left(A\right)^T$

(e) $\left(A_1 A_2 A_3 ... A_{n-1} A_n\right)^T = A_n^T A_{n-1}^T .... A_3^T A_2^T A_1^T$

The transpose of

$$[A]=\begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

The transpose of [A] is $[A]^T = \begin{bmatrix} 25 & 5 & 6 \\ 20 & 10 & 16 \\ 3 & 15 & 7 \\ 2 & 25 & 27 \end{bmatrix}$

Note, the transpose of a row vector is a column vector and the transpose of a column vector is a row vector.
Also, note that the transpose of a transpose of a matrix is the matrix itself, that is, $([A]^T)^T = [A]$ Also, $(A+B)^T = A^T + B^T$; $(cA)^T = cA^T$.

1.3.10 Traces of a matrix

- The trace of a $n \times n$ matrix [A] is the sum of the diagonal entries of [A], that is,

$$tr[A]=\sum_{i=1}^{n} a_{ii}$$

Example:
The trace of

$$[A]=\begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix}$$ is $tr[A]=\sum_{i=1}^{3} a_{ii}$ = (15) + (-4) + (6) =17

1.4 Determinant of a matrix:

- The determinant of a square matrix is a single unique real number corresponding to a matrix. For a matrix [A], determinant is denoted by |A| or det(A). So do not use [A] and |A| interchangeably.
- For a 2×2 matrix,

$$[A]=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

det (A) = $a_{11}\, a_{22} - a_{12}\, a_{21}$

Career Avenues

1.4.1 Calculation of determinant of any square matrix

- Let [A] be n $\times$ n matrix. The minor of entry a_{ij} is denoted by M_{ij} and is defined as the determinant of the $(n-1)\times(n-1)$ submatrix of [A], where the submatrix is obtained by deleting the ith row and j^{th} column of the matrix [A]. The determinant is then given by

$$\det(A)=\sum_{j=1}^{n}(-1)^{i+j}\,a_{ij}\,M_{ij}\ \ for\ any\ i=1,2,...,n$$

or

$$\det(A)=\sum_{i=1}^{n}(-1)^{i+j}\,a_{ij}\,M_{ij}\ \ for\ any\ j=1,2,...,n$$

coupled with that $\det(A)=a_{11}\ for\ a\ 1\times1 matrix\ [A]$, as we can always reduce the determinant of a matrix to determinants of 1×1 matrices. The number $(-1)^{i+j}M_{ij}$ is called the cofactor of a_{ij} and is denoted by c_{ij}. The above equation for the determinant can then be written as

$$\det(A)=\sum_{j=1}^{n}a_{ij}\,C_{ij}\ \ for\ any\ i=1,2,...,n$$

or

$$\det(A)=\sum_{i=1}^{n}a_{ij}\,C_{ij}\ \ for\ any\ j=1,2,...,n$$

The only reason why determinants are not generally calculated using this method is that it becomes computationally intensive.

Example:

Find the determinant of $[A]=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$

Method 1 (for the example given above)

$$\det(A)=\sum_{j=1}^{3}(-1)^{i+j}\,a_{ij}\,M_{ij}\ \ for\ any\ i=1,2,3$$

Let i = 1 in the formula

$$\det(A)=\sum_{j=1}^{3}(-1)^{1+j}\,a_{1j}\,M_{1j}$$

$$=(-1)^{1+1}a_{11}M_{11}+(-1)^{1+2}a_{12}M_{12}+(-1)^{1+3}a_{13}M_{13}$$

$$=a_{11}M_{11}-a_{12}M_{12}+a_{13}M_{13}$$

$$M_{11}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}8 & 1\\ 12 & 1\end{vmatrix}$$

= - 4

Career Avenues

$$M_{12}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}64 & 1\\ 144 & 1\end{vmatrix}=-80$$

$$M_{13}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}64 & 8\\ 144 & 12\end{vmatrix}=-284$$

$$\det(A)=a_{11}M_{11}-a_{12}M_{12}+a_{13}M_{13}$$

$$=25(-4)-5(-80)+1(-384)$$

$= -100 + 400 - 384$

$= -84$

Also for i = 1

$$\det(A)=\sum_{j-1}^{3}a_{1j}C_{1j}$$

$$C_{11}=(-1)^{1+1}M_{11}$$

$= M_{11}$

$= -4$

$$C_{12}=(-1)^{1+2}M_{12}$$

$= -M_{12}$

$= 80$

$$C_{13}=(-1)^{1+3}M_{13}$$

$= M_{13}$

$= -284$

$$\det(A)=a_{11}C_{11}+a_{21}C_{21}+a_{31}C_{31}$$

$= (25)(-4) + (5)(80) + (1)(-284)$

$= -100 + 400 - 384$

$= -84$

Method 2:

$$\det(A)=\sum_{i=1}^{3}(-1)^{i+j}a_{ij}M_{ij}$$ for any j = 1,2,3

Let j = 2 in the formula

$$\det(A)=\sum_{i=1}^{3}(-1)^{i+2}a_{i2}M_{i2}$$

$$=(-1)^{1+2}a_{12}M_{12}+(-1)^{2+2}a_{22}M_{22}+(-1)^{3+2}a_{32}M_{32}$$

$$=-a_{12}M_{12}+a_{22}M_{22}-a_{32}M_{32}$$

$$M_{12}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}64 & 1\\ 144 & 1\end{vmatrix}$$

$= -80$

$$M_{22}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}25 & 1\\ 144 & 1\end{vmatrix}$$

$= -119$

Career Avenues

$$M_{32} = \begin{vmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{vmatrix} = \begin{vmatrix} 25 & 1 \\ 64 & 1 \end{vmatrix}$$

= -39

$\det(A) = -a_{12} M_{12} + a_{22} M_{22} - a_{32} M_{32}$

= -5 (-80) + 8 (-119) -12 (-39)

= 400 – 952 + 468

= - 84

In terms of cofactors for j = 1,

$\det(A) = \sum_{i=1}^{3} a_{i2} C_{12}$

$C_{12} = (-1)^{1+2} M_{12}$

= - M_{12}

= 80

$C_{22} = (-1)^{2+2} M_{22}$

= M_{22}

= - 119

$C_{32} = (-1)^{3+2} M_{32}$

= - M_{32}

= 39

$\det(A) = a_{12} C_{12} + a_{22} C_{22} + a_{32} C_{32}$

= (5) (80) + (8) (-119) + (12) (39)

= 400 – 952 + 468

= -84

1.4.2 Theorems related to Determinant of Square Matrix:

Theorem 1: If a row or a column in a $n \times n$ matrix [A] is zero, then det (A) = 0.

Theorem 2: Let [A] be a $n \times n$ matrix. If a row is proportional to another row, then det (A) = 0.

Theorem 3: Let [A] be a $n \times n$ matrix. If a column is proportional to another column, then det (A) = 0.

Theorem 4: Let [A] be a $n \times n$ matrix. If a column or row is multiplied by k to result in matrix k, then det (B) = k det (A)

Theorem 5: Let [A] be a $n \times n$ upper or lower triangular matrix, then $\det(B) = \prod_{i=1}^{n} a_{ii}$.

Career Avenues

1.4.3 Properties of determinant of a square matrix

(a) $|A|$ exists $\Leftrightarrow$ A is a square matric
(b) $|AB| = |A|\,|B|$
(c) $|A^T| = |A|$
(d) $|kA| = k^n |A|$, if A is a square matrix of order n.
(e) If A and B are square matrices of same order then $|AB| = |BA|$
(f) If A is a skew symmetric matric of odd order the $|A| = 0$
(g) If $A = \text{diag}(a_1, a_2 \ldots a_n)$ then $|A| = a_1 a_2 \ldots a_n$.
(h) $|A|^n = |A^n|$, $n \in N$.
(i) If $|A| = 0$, then matrix is called singular

Example:
What is the determinant of

$$[A]=\begin{bmatrix}0 & 2 & 6 & 3\\ 0 & 3 & 7 & 4\\ 0 & 4 & 9 & 5\\ 0 & 5 & 2 & 1\end{bmatrix}$$

Solution:
Since one of the columns (first column in the above example) of [A] is a zero, det (A) = 0 .

Example
What is the determinant of

$$[A]=\begin{bmatrix}2 & 1 & 6 & 4\\ 3 & 2 & 7 & 6\\ 5 & 4 & 2 & 10\\ 9 & 5 & 3 & 18\end{bmatrix}$$

Solution
det (A) is zero because the fourth column

$\begin{bmatrix}4\\6\\10\\18\end{bmatrix}$ **is 2 times the first column** $\begin{bmatrix}2\\3\\5\\9\end{bmatrix}$

Example
If the determinant of

$$[A]=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$$

is -84, then what is the determinant of

$$[B]=\begin{bmatrix}25 & 10.5 & 1\\ 64 & 16.8 & 1\\ 144 & 25.2 & 1\end{bmatrix}$$

Solution
Since the second column of [B] is 2.1 times the second column of [A]
det (B) = 2.1 det (A)

Career Avenues

= (2.1) (-84)
= - 176.4

Example
Given the determinant of

$$[A]=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$$

is -84, what is the determinant of

$$[B]=\begin{bmatrix}25 & 5 & 1\\ 0 & -4.8 & -1.56\\ 144 & 12 & 1\end{bmatrix}$$

Solution
Since [B] is simply obtained by subtracting the second row of [A] by 2.56 times the first row of [A],
det (B) = det (A)
= - 84

Example
What is the determinant of

$$[A]=\begin{bmatrix}25 & 5 & 1\\ 0 & -4.8 & -1.56\\ 0 & 0 & 0.7\end{bmatrix}$$

Solution
Since [A] is an upper triangular matrix

$$\det(A)=\prod_{i=1}^{3}a_{ii}$$

$$=(a_{11})(a_{22})(a_{33})$$

= (25) (-4.8) (0.7)
=-84

1.5 Systems of Linear equations: -

Consider the following sets of equations

$$a_{11}x+a_{12}y+a_{13}z=b_1$$
$$a_{21}x+a_{22}y+a_{23}z=b_2$$
$$a_{31}x+a_{32}y+a_{33}z=b_3$$

The system of equation can be written as

Ax = B

Where

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}\quad x=\begin{bmatrix}x\\ y\\ z\end{bmatrix}\qquad B=\begin{bmatrix}b_1\\ b_2\\ b_3\end{bmatrix}$$

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The above system of equation is Consistent only if it has at least one solution exit & if no solution exists then it will be Inconsistent.

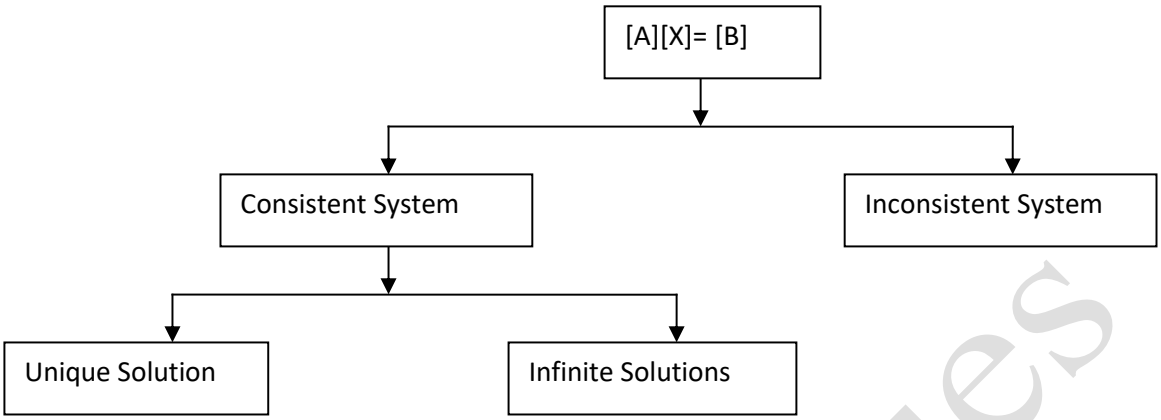


Figure: Consistent and inconsistent system of equations flow chart.

1.6 Rank of a matrix

- The rank of a matrix is defined as the order of the largest square sub matrix whose determinant is not zero.

OR

- The Augmented Matrix is defined as

$$[A:B]=\begin{bmatrix} a_{11} & a_{12} & a_{13} & :b_1 \\ a_{21} & a_{22} & a_{23} & :b_2 \\ a_{31} & a_{32} & a_{33} & :b_3 \end{bmatrix}$$

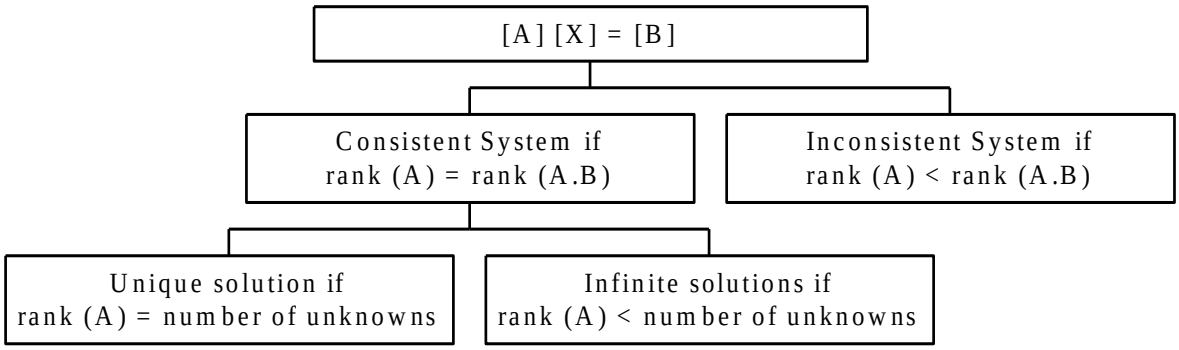
- ***If the rank of matrix [A] is equal to the rank of augmented Matrix [A:B], then the system of equations AX=B is consistent and has either unique or infinite solution.***
- ***If the rank of matrix [A] is not equal to Rank of Augmented matrix [A:B] then system will be Inconsistent and has no solution.***

1.7 Consistent and inconsistent system of equations: -

- A system of equations is consistent if there is a solution and it is inconsistent if there is no solution. However, a consistent system of equations does not mean a unique solution, that is, a consistent system of equations may have a unique solution or infinite solutions. If [A][X]=[B] is a system of Linear Equation.

  Where [A] is square matrix, [A:B] is a augmented Matrix and [B] is a column Matrix then

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Consistency check for Non-Homogenous equation [A][X] = [B]

- When |A|≠0, then system of equations has unique solution.
- When |A| = 0 then system of equations has either Infinite many solutions or no solution

(i) If |A|=0 & rank of [A] = rank of [A: B], then system of equations have Infinite solutions.

(ii) If |A|=0 & rank of [A] ≠ rank of [A:B], then system of equations have no solution.

Consistency check for Homogenous equation [A][X] = [0]

Homogenous equations are always consistent, so only two conditions occurs: -

- When |A| = 0, then Infinite many solutions or Non-Trivial solution
- When |A|≠0, then unique solution or trivial solution.

Example:
What is the rank of

$$[A]=\begin{bmatrix}3 & 1 & 2\\ 2 & 0 & 5\\ 1 & 2 & 3\end{bmatrix}?$$

Solution:
The largest square sub matrix possible is of order 3 and that is [A] itself. Since $\det(A)=-23\neq 0$ the rank of [A] = 3.

Example:
What is the rank of

$$[A]=\begin{bmatrix}3 & 1 & 2\\ 2 & 0 & 5\\ 5 & 1 & 7\end{bmatrix}?$$

Solution:
The largest square submatrix of [A] is of order 3 and that is [A] itself. Since det (A) = 0, the rank of [A] is less than 3. The next largest square submatrix would be a 2×2 matrix. One of the square submatrices of [A] is

$$[B]=\begin{bmatrix}3 & 1\\ 2 & 0\end{bmatrix}$$

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and $\det(B) = -2 \neq 0$ Hence the rank of [A] is 2. There is no need to look at other 2×2 submatrices to establish that the rank of [A] is 2.

Example:
Use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

is a consistent or inconsistent system of equations?

Solution:
The coefficient matrix is

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and the right hand side vector is

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & \vdots & 106.8 \\ 64 & 8 & 1 & \vdots & 177.2 \\ 144 & 12 & 1 & \vdots & 279.2 \end{bmatrix}$$

Since there are no square submatrices of order 4 as [B] is a 3×4 matrix, the rank of [B] is at most 3. So let us look at the square submatrices of [B] of order 3; if any of these square submatrices have determinant not equal to zero, then the rank is 3. For example, a submatrix of the augmented matrix [B] is

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

has det (D) = -84 $\neq$ 0.
Hence the rank of the augmented matrix [B] is 3. Since [A] = [D], the rank of [A] is 3. Since the rank of the augmented matrix [B] equals the rank of the coefficient matrix [A], the system of equations is consistent.

Example:
Use the concept of rank of matrix to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

is consistent or inconsistent?

Solution:
The coefficient matrix is given by

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

and the right hand side

Career Avenues

$$[C]=\begin{bmatrix}106.8\\177.2\\284.0\end{bmatrix}$$

The augmented matrix is

$$[B]=\begin{bmatrix}25 & 5 & 1:106.8\\64 & 8 & 1:177.2\\89 & 13 & 2:284.0\end{bmatrix}$$

Since there are no square submatrices of order 4 as [B] is a 4×3 matrix, the rank of the augmented [B] is at most 3. So let us look at square submatrices of the augmented matrix [B] of order 3 and see if any of these have determinants not equal to zero. For example, a square submatrix of the augmented matrix [B] is

$$[D]=\begin{bmatrix}25 & 5 & 1\\64 & 8 & 1\\89 & 13 & 2\end{bmatrix}$$

has det (D) = 0. This means, we need to explore other square submatrices of order 3 of the augmented matrix [B] and find their determinants.

That is,

$$[E]=\begin{bmatrix}5 & 1 & 106.8\\8 & 1 & 177.2\\13 & 2 & 284.0\end{bmatrix}$$

det (E) = 0

$$[F]=\begin{bmatrix}25 & 5 & 106.8\\64 & 8 & 177.2\\89 & 13 & 284.0\end{bmatrix}$$

det (F) = 0

$$[G]=\begin{bmatrix}25 & 1 & 106.8\\64 & 1 & 177.2\\89 & 2 & 284.0\end{bmatrix}$$

det (G) = 0

All the square submatrices of order 3×3 of the augmented matrix [B] have a zero determinant. So the rank of the augmented matrix [B] is less than 3. Is the rank of augmented matrix [B] equal to 2?. One of the 2×2 submatrices of the augmented matrix [B] is

$$[H]=\begin{bmatrix}25 & 5\\64 & 8\end{bmatrix}$$

and

det (H) = - 120 $\neq 0$

So the rank of the augmented matrix [B] is 2.

Now we need to find the rank of the coefficient matrix [B].

$$[A]=\begin{bmatrix}25 & 5 & 1\\64 & 8 & 1\\89 & 13 & 2\end{bmatrix}$$

and

det (A) = 0

So the rank of the coefficient matrix [A] is less than 3. A square submatrix of the coefficient matrix [A] is

$$[J]=\begin{bmatrix}5 & 1\\8 & 1\end{bmatrix}$$

det (J) = -3 $\neq 0$

So the rank of the coefficient matrix [A] is 2.

Career Avenues

Hence, rank of the coefficient matrix [A] equals the rank of the augmented matrix [B]. So the system of equations [A] [X] = [C] is consistent.

Example:
Use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 280.0 \end{bmatrix}$$

is consistent or inconsistent.

Solution:
The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & :106.8 \\ 64 & 8 & 1 & :177.2 \\ 89 & 13 & 2 & :280.0 \end{bmatrix}$$

Since there are no square submatrices of order 4×4 as the augmented matrix [B] is a 4×3 matrix, the rank of the augmented matrix [B] is at most 3. So let us look at square submatrices of the augmented matrix (B) of order 3 and see if any of the 3×3 submatrices have a determinant not equal to zero. For example, a square submatrix of order 3×3 of [B]

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

det(D) = 0
So it means, we need to explore other square submatrices of the augmented matrix [B]

$$[E] = \begin{bmatrix} 5 & 1 & 106.8 \\ 8 & 1 & 177.2 \\ 13 & 2 & 280.0 \end{bmatrix}$$

det (E0 $\neq 12.0 \neq 0$)
So the rank of the augmented matrix [B] is 3.
The rank of the coefficient matrix [A] is 2 from the previous example.
Since the rank of the coefficient matrix [A] is less than the rank of the augmented matrix [B], the system of equations is inconsistent. Hence, no solution exists for [A] [X] [C].

If a solution exists, how do we know whether it is unique?
In a system of equations [A] [X] = [C] that is consistent, the rank of the coefficient matrix [A] is the same as the augmented matrix [A|C]. If in addition, the rank of the coefficient matrix [A] is same as the number of unknowns, then the solution is unique; if the rank of the coefficient matrix [A] is less than the number of unknowns, then infinite solutions exist.

Example:
We found that the following system of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.0 \end{bmatrix}$$

is a consistent system of equations. Does the system of equations have a unique solution or does it have infinite solutions?

Career Avenues

Solution:
The coefficient matrix is

$$[A]=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$$

and the right hand side is

$$[C]=\begin{bmatrix}106.8\\ 177.2\\ 279.2\end{bmatrix}$$

While finding out whether the above equations were consistent in an earlier example, we found that the rank of the coefficient matrix (A) equals rank of augmented matrix $[A\vdots C]$ equals 3.
The solution is unique as the number of unknowns = 3 = rank of (A).

1.8 Inverse of a matrix: -

If [A] is any square matrix of order (n x n) then, $[A]^{-1}$ will be the Inverse of [A].

Such that $[A]\,[A]^{-1} = [I]$

$$\text{And } A^{-1} = \frac{adj[A]}{|A|}$$

Where adj[A] = Adjoint of Matrix [A]

|A| = Determinant of Matrix [A]

[I]= Unity Matrix

1.8.1 Adjoint of Matrix

Adjoint of Matrix [A] is obtained by: -

- First replace all elements by their respective Co-Factors
- Take transpose of the matrix obtained at step (a), the resultant will be the

Adjoint of [A].

$$[A]^{-1} = \frac{1}{set(A)}adj(A)$$

where

$$adj(A)=\begin{bmatrix}C_{11} & C_{12} & \dots C_{1n}\\ C_{12} & C_{22} & C_{2n}\\ \vdots & & \\ C_{ml} & C_{n2} & \cdots C_{nn}\end{bmatrix}$$ where C_{ij} are the cofactors of a_{ij}. The matrix

$$\begin{bmatrix}C_{11} & C_{12} & \cdots C_{1n}\\ C_{21} & C_{22} & \cdots C_{2n}\\ \vdots & & \vdots\\ C_{n1} & \cdots & \cdots C_{nn}\end{bmatrix}$$ itself is called the matrix of cofactors from [A].

Career Avenues

Example:
Find the inverse of

$$[A]=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$$

Solution:
From previous examples, we found
det (A) = - 84
Next we need to find the adjoint of [A]. The cofactors of A are found as follows.
The minor of entry a_{11} is

$$M_{11}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}8 & 1\\ 12 & 1\end{vmatrix}=-4$$

The cofactors of entry a_{11} is
$C_{11} = (-1)^{1+1}M_{11}$
$=M_{11}$
$= - 4$
The minor of entry a_{12} is

$$M_{12}=\begin{vmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{vmatrix}=\begin{vmatrix}64 & 1\\ 144 & 1\end{vmatrix}$$

= -80
The cofactor of entry a_{12} is
$C_{12} = (-1)^{1+2} M_{12}$
$=-M_{12}$
$= - M_{12}$
$= - (-80)$
$=80$
Similarly

$$C_{13}=-384$$

$C_{13} = - 384$
$C_{21} = 7$
$C_{22} = -119$
$C_{23} = 420$
$C_{31} = -3$
$C_{32} = 39$
$C_{33} = -120$
Hence the matrix of cofactors of [A] is

$$[C]=\begin{bmatrix}-4 & 80 & -384\\ 7 & -119 & 420\\ -3 & 39 & -120\end{bmatrix}$$

The adjoint of matrix [A] is $[C]^T$,

$$adj(A)=[C]^T$$

Adj (A) = $[C]^T$

$$=\begin{bmatrix}-4 & 7 & -3\\ 80 & -119 & 39\\ -384 & 420 & -120\end{bmatrix}$$

Hence

Career Avenues

$$[A]^{-1} = \frac{1}{\det(A)} adj(A)$$

$$= \frac{1}{-84}\begin{bmatrix} -4 & 7 & -3 \\ 80 & -119 & 39 \\ -384 & 420 & -120 \end{bmatrix}$$

$$= \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

1.8.2 Properties of Inverse of Matrix:

(a) $A\ (adj\ A) = |A|\ I_n = (adj\ A)A$
(b) $|adj\ A| = |A|^{n-1}$
(c) $adj\ (adj\ A) = |A|^{n-2}\ A$
(d) $|adj\ (adj\ A)| = |A|^{(n-1)2}$
(e) $adj\ (A^T) = (adj\ A)^T$
(f) $adj\ (AB) = (adj\ B)\ (adj\ A)$
(g) $adj\ (A^m) = (adj\ A)^m,\ m \in N$
(h) $adj\ (kA) = k^{n-1}\ (adj\ A),\ k \in R$

1.8.3 Properties of Adjoint of Matrix

(a) $(A^T)^{-1} = (A^{-1})^T$
(b) $(AB)^{-1} = B^{-1}A^{-1}$
(c) $(A^k)^{-1} = (A^{-1})^k,\ k \in N$
(d) $adj\ (A^{-1}) = (adj\ A)^{-1}$
(e) $|A^{-1}| = \frac{1}{|A|} = |A|^{-1}$
(f) If $A = diag\ (a_1, a_2, \ldots a_n)$, then $A^{-1} = diag\ (a_1^{-1}, a_2^{-1}, \ldots, a_n^{-1})$
(g) $AB = AC \Rightarrow B = C$ if $|A| \neq 0$,

1.9 Eigen values and Eigen vectors

What is the general definition of eigenvalues and eigenvectors of a square matrix?

If [A] is a n $\times$ n matrix, then $[X] \neq \bar{0}$ is an eigenvector of [A] if $[A][X] = \lambda[X]$

where λ is a scalar and $[X] \neq 0$.

The scalar λ is called the eigenvalue of [A] and [A] is called the eigenvector corresponding to the eigenvalue λ.

1.9.1 Method for finding Eigen values of a square matrix

To find the eigenvalues of a n $\times$ n matrix [A], we have $[A][X] = \lambda[X]$

$[A][X] - \lambda[X] = 0$

$[A][X] - \lambda[I][X] = 0$

Career Avenues

$([A]-[\lambda][I])[X]=0$

Now for the above set of equations to have a nonzero solution,

$\det([A]-\lambda[I])=0$

The above equation is called the characteristic equation of [A].

This left-hand side can be expanded to give a polynomial in λ solving the above equation would give us values of the eigenvalues.

For a [A] $n\times n$ matrix, the characteristic polynomial of A is of degree n as follows

$\det([A]-\lambda[I])=0$

giving

$\lambda^n+c_1\lambda^{n-1}+c_2\lambda^{n-2}+---+c_n=0$

Hence. this polynomial has n roots.

Example:
Find the eigenvalues of the matrix

$$[A]=\begin{bmatrix}3 & -1.5\\ -0.75 & 0.75\end{bmatrix}$$

Solution:

$$[A]-\lambda[I]=\begin{bmatrix}3-\lambda & -1.5\\ -0.75 & 0.75-\lambda\end{bmatrix}$$

$$\det([A]-\lambda[I])=(3-\lambda)(0.75-\lambda)-(-0.75)(-1.5)=0$$

$2.25 - 0.75\ \lambda - 3\lambda + \lambda^2 - 1.125 = 0$

$\lambda^2 - 3.75\lambda + 1.125 = 0$

$$\lambda=\frac{-(-3.75)\pm\sqrt{(-3.75)^2-4(1)(1/125)}}{2(1)}$$

$$=\frac{3.75\pm3.092}{2}=3.421,0.3288$$

So, the eigenvalues are 3.421 and 0.3288.

Example:
Find the eigenvectors of

$$A=\begin{bmatrix}3 & -1.5\\ -0.75 & 0.75\end{bmatrix}$$

Solution:
The eigenvalues have already been found in Example 1 as
$\lambda_1=3.421,\lambda_2=0.3288$
Let

$$[X]=\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

be the eigenvector corresponding to
$\lambda_1=3.421$
Hence
$([A]-\lambda_1[I])[X]=0$

Career Avenues

$$\left\{\begin{bmatrix}3 & -1.5\\ -0.75 & 0.75\end{bmatrix}-3.421\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right\}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=0$$

$$\begin{bmatrix}-0.421 & -1.5\\ -0.75 & -2.671\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

If

$x_1 = s$

then

$-0.421s - 1.5x_2 = 0$

$x_2 = -0.2808s$

The eigenvector corresponding to $\lambda_1 = 3.421$ then is

$$[X]=\begin{bmatrix}s\\ -0.2808s\end{bmatrix}$$

$$=s\begin{bmatrix}1\\ -0.2808\end{bmatrix}$$

The eigenvector corresponding to

$$\lambda_1 = 3.421is\begin{bmatrix}1\\ -0.2808\end{bmatrix}$$

Similarly, the eigenvector corresponding to

$$\lambda_2 = 0.3288is\begin{bmatrix}1\\ 1.781\end{bmatrix}$$

Example:

Find the eigenvalues and eigenvectors of

$$[A]=\begin{bmatrix}1.5 & 0 & 1\\ -0.5 & 0.5 & -0.5\\ -0.5 & 0 & 0\end{bmatrix}$$

Solution:

The characteristic equation is given by

$$\det([A]-\lambda[I])=0$$

$$\det\begin{bmatrix}1.5-\lambda & 0 & 1\\ -0.5 & 0.5-\lambda & -0.5\\ -0.5 & 0 & -\lambda\end{bmatrix}=0$$

$$(1.5-\lambda)\left[(0.5-\lambda)-(-0.5)(0)\right]+(1)\left[(-0.5)(0)-(-0.5)(0.5-\lambda)\right]=0$$

$$-\lambda^3+2\lambda^2-1.25\lambda=0$$

The roots of the above equation are

$\lambda = 0.5, 0.5, 1.0$

Note that there are eigenvalues that are repeated. Since there are only two distinct eigenvalues, there are only two eigenspaces. But, corresponding to λ = 0.5 there should be two eigenvectors that form a basis for the eigenspace.

Career Avenues

To find the eigenspaces, let

$$[X]=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Given

$$\left[(A-\lambda I)\right][X]=0$$

then

$$\begin{bmatrix} 1.5-\lambda & 0 & 1 \\ -0.5 & 0.5-\lambda & -0.5 \\ -0.5 & 0 & -\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 0.5$,

$$\begin{bmatrix} 1 & 0 & 1 \\ -0.5 & 0 & -0.5 \\ -0.5 & 0 & -0.5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$x_1 = -a, x_2 = b x_3 = a$

So

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} -a \\ b \\ a \end{bmatrix}$$

$$=\begin{bmatrix} a \\ 0 \\ -a \end{bmatrix}+\begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix}$$

$$=a\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}+b\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

So the vectors $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ form a basis for the eigenspace for the eigenvalue $\lambda = 0.5$.

For $\lambda = 1$,

$$\begin{bmatrix} 0.5 & 0 & 1 \\ -0.5 & -0.5 & -0.5 \\ -0.5 & 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$x_1 = a, x_2 = -0.5a, x_3 = -0.5a$

The eigenvector corresponding to $\lambda = 1$ is

$$\begin{bmatrix} a \\ -0.5a \\ -0.5a \end{bmatrix}=a\begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Hence the vector $\begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$ is a basis for the eigenspace for the eigenvalue of $\lambda = 1$.

Career Avenues

1.9.2 Theorems related to Eigen values and Eigen vectors

Theorem 1: If [A] is a n × n triangular matrix – upper triangular, lower triangular or diagonal, the eigenvalues of [A] are the diagonal entries of [A].

Theorem 2: $\lambda = 0$ is an eigenvalue of [A] if [A] is a singular (noninvertible) matrix.

Theorem 3: [A] and $[A]^T$ have the same eigenvalues.

Theorem 4: Eigenvalues of a symmetric matrix are real.

Theorem 5: Eigenvectors of a symmetric matrix are orthogonal, but only for distinct eigenvalues.

Theorem 6: Determinant of [A] i.e |A| is equal to the product of the absolute values of the eigenvalues of [A].

Theorem 7: Sum of diagonal elements of matrix is equals to sum of eigen values of matrix

If $\lambda_1, \lambda_2 .....\lambda_n$ are the n eigen values of A, then eigenvalues of

(1) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2} ..... \frac{1}{\lambda_n}$

(2) kA are $k\lambda_1, k\lambda_2 .....k\lambda_n$ when k is any scalar

(3) A^m are $\lambda_1^n, \lambda_2^m, ...\lambda_2^m$

1.10 Caley Hamilton theorem

Every square matrix satisfies its characteristics equation

$$\det([A] - \lambda[I]) = 0$$

i.e. if [A] is a square matrix of order n, then Matrix [A] will satisfy its characteristics equation by putting (λ=A) in $\det([A] - \lambda[I]) = 0$

Example:
What are the eigenvalues of

$$[A] = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 7 & 3 & 0 & 0 \\ 9 & 5 & 7.5 & 0 \\ 2 & 6 & 0 & -7.2 \end{bmatrix}$$

Solution:
Since the matrix [A] is a lower triangular matrix, the eigenvalues of [A] are the diagonal elements of [A]. The eigenvalues are
$\lambda_1 = 6, \lambda_2 = 3, \lambda_3 = 7.5, \lambda_4 = 7.2$

Example:
One of the eigenvalues of

$$[A] = \begin{bmatrix} 5 & 6 & 2 \\ 3 & 5 & 9 \\ 2 & 1 & -7 \end{bmatrix}$$

Career Avenues

is zero. Is [A] invertible?

Solution:
$\lambda = 0$ is an eigenvalue of [A] , that implies [A] is singular and is not invertible.

Example:
Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$\lambda_1 = -1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$

Then find the eigenvalues of [B] if

$$[B] = \begin{bmatrix} 2 & 3.5 & 8 \\ -3.5 & 5 & 1 \\ 6 & 2 & 8.5 \end{bmatrix}$$

Solution:
Since $[B] = [A]^T$, the eigenvalues of [A] and [B] are the same. Hence eigenvalues of [B] also are

$\lambda_1 = 1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$

Example:
Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$\lambda_1 = 1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$

Calculate the magnitude of the determinant of the matrix.

Solution:
Since

$$|\det[A]| = |\lambda_1||\lambda_2||\lambda_3|$$

$$= |-1.547||12.33||4.711|$$

= 89.88

Career Avenues

SET-1

1. The Rank of the given Matrix is

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

a) 1 b) 2 c) 3 d) 4

2. If the determinant of Matrix [A] is 7, i.e |A|=7, Then the determinant of inverse of Matrix [2A] is……. ,Where [A] is given by

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & b & c \end{bmatrix}$$

a) 7/2 b) 1/56 c) 1/49 d)1/14

3. The determinant of matrix [A] is……,
Where [A] is given by

$$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

a) 0 b) 1 c) -1 d) 2

4. The determinant of the matrix [A] is……., Where [A] is given by

$$\begin{bmatrix} 5 & 3 & 2 \\ 1 & 2 & 6 \\ 3 & 5 & 10 \end{bmatrix}$$

a) -50 b) -20 c) -73 d) -28

5. It is given that the Matrix [A] is Singular, then λ =……., where [A] is given by

$$\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -2 & \lambda \end{bmatrix}$$

a) 0 b) 1 c) -2 d) -1

6. The Adjoint of the Matrix [A] is....., Where [A] is given by

$$\begin{bmatrix} 1 & -4 \\ 0 & 2 \end{bmatrix}$$

a) $\begin{bmatrix} 2 & 4 \\ 1 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}$ c) $\begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$ d) $\begin{bmatrix} 1 & 0 \\ 4 & 2 \end{bmatrix}$

Career Avenues

7. The Value of (a+b) will be equal to……., if Matrix [A] and its Inverse is given by

$$A=\begin{bmatrix}2 & -01\\ 0 & 3\end{bmatrix}\ A^{-1}=\begin{bmatrix}\frac{1}{2} & a\\ 0 & b\end{bmatrix}$$

a) 2/15 b) 3/17 c) 7/20 d) 13/17

8. Matrix [X] is an orthogonal Matrix, the value of |B|=……, Where Matrix [X] is given by

$$\begin{bmatrix}A & B\\ C & 0\end{bmatrix}$$

a) 0 b) 1 c) ½ d) 2

9. A and B are Matrices of order 4x4, where A=5B and |A|= α |B|, then the Value of α is…….

a) 10 b) 625 c) 500 d) 325

10. For the following systems of equations

$$x-4y+7z=14$$
$$3x+8y-2z=13$$
$$7x-8y+26z=5$$

The nature of solution will be

a) Unique solution b) Infinite many soution

c) No solution d) Two Solutions

11. Given matrix A as follows, then A^2 is equal to

$$A=\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

a) 3A b) 2A c) 4A d) 0.5 A

12. Which of following is true?

a) Inverse of skew – symmetric matrix is symmetric

b) Inverse of lower – triangular is upper triangular but vice-versa is not possible

c) Inverse of upper – triangular is lower – triangular but vice-versa is not possible

d) Inverse of diagonal matrix is always diagonal

13. The system of eqautions given below has a non-zero solution, If λ=…….

$$3x-y+z=0$$
$$15x-6y+5z=0$$
$$\lambda x-2y+2z=0$$

a) 2 b) 4 c) 6 d) 8

Career Avenues

14. It is given that the following set of equations is Inconsistent, then λ=……

$x+y+z=6$

$x+2y+3z=10$

$x+2y+\lambda z=12$

a) 2 b) -2 c) 3 d) -3

15. The Eigen Values of the given matrix are

$\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$

a) 2 and 1 b) 4 and 1 c) 0 and 5 d)4 and 0

16. The Eigen Values of the given matrix are

$\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$

a)1, 0, 0 b) 0,1,1 c) 2,-1,0 d) 2,1,1

17. One of the eigen value of the matrix is 3, then the other two eigen values of matrix are

$A=\begin{bmatrix} 2 & -2 & 3 \\ -2 & -1 & 6 \\ 1 & 2 & 0 \end{bmatrix}$

a) -3 and 5 b) 3 and -5 c) 3 and 3 d) 5 and 5

18. If 1,3 and 4 are the eigen values of matrix [A], then the determinant of the Matrix |A| =…….

a) 4 b) 8 c) 12 c) 15

19. The eigen values of non-singular matrix are 2, 4, and -4, then the eigen value of Inverse of [A] are

a) ½, ¼ , -¼ b) 2, 4,-4

c) 2|A|, 4|A|, -4|A| d) 2/|A , 4/|A|, -4/|A|

20. The eigen value of a square matrix A are 1, -2, and 3. The eigen values of the matrix [2A] will be

a) 1, -2 and 3 b) 2, -4 and 6 c) 1, 4 and 6 d) -2, 2 and 6

21. The eigen value of a square matrix [A] given below are 1, and 5. The eigen values of the matrix [A][A] = Square of [A] will be……..

a) 1 and 5 b) 1 and 25 c) 2 and 10 d) 1/5 and 1/25

22. Which one of the following is the eigen vector the matrix [A] , where [A] is given by

$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$

a) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ b) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ c) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ d) $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

23. Eigen Values of a matrix [A] are 1,-1, and 0, then the determinant of $I+A^{100}$ is

a) 2 b) 6 c) 8 d)4

Career Avenues

24. The eigen values of a Matrix [P] are 1,-2, and 3 then the value of Inverse of [P] is

a) $P^{-1}=\frac{1}{6}\left(5I+2P-P^2\right)$

b) $P^{-1}=\frac{1}{6}\left(5I-2P-P^2\right)$

c) $P^{-1}=\frac{1}{6}\left(5I-2P+P^2\right)$

d) $P^{-1}=\left(5I+2P+P^2\right)$

25. The eigen values of Matrix [A] are 1 and 3, then A^3 is equal to

a) 12A- 13I
b) 13A-12I
c) 12(A-I)
d) 13(A-I)

Career Avenues

Solutions (Set-1)

1. Ans: (c) It's a fourth order matrix and determinant of Matrix is 0, So highest order Non-Zero sub matrix is 3,

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & -8 \\ 0 & 0 & -3 \end{bmatrix} = (1)(-4)(-3) = 12 \neq 0$$

So Rank is 3.

2. Ans: b) By using the property $|2A| = 2^3|A|$, where n=3 is the order of Matrix [A]

$$\det(2A)^{-1} = \frac{1}{|2A|} = \frac{1}{2^n|A|}$$

Since A is a 3 x 3 matric, therefore n = 3 and we have $|A| = 7$

$$\det(2A)^{-1} = \frac{1}{2^3|A|} = \frac{1}{8\times 7} = \frac{1}{56}$$

3. Ans: (c)

We have $$A = \begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

Expanding cofactor of a_{34} $$|A| = -1\begin{bmatrix} 0 & 1 & 0 \\ -1 & 1 & 1 \\ 1 & -2 & 0 \end{bmatrix}$$

$$= -(0 - 1(0-1) + 0)$$

=-1

4. Ans: (d)

$$\begin{bmatrix} 5 & 3 & 2 \\ 1 & 2 & 6 \\ 3 & 5 & 10 \end{bmatrix} = 5(20-30) - 3(10-18) + 2(5-6)$$

$$= -50 + 24 - 2 = -28$$

5. Ans: (c) here it is given that [A] is a singular matrix so |A|=0

Matric A is singular is $|A| = 0$

Or $$\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -2 & \lambda \end{bmatrix} = 0$$

Or $$-(-1)\begin{bmatrix} 1 & -2 \\ -2 & \lambda \end{bmatrix} + 2\begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} + 0\begin{bmatrix} 0 & 3 \\ -2 & \lambda \end{bmatrix} = 0$$

Or $$(\lambda - 4) + 2(3) = 0$$

Or $$\lambda = -2$$

Career Avenues

6. Ans: (b)

$$A=\begin{bmatrix}1 & -4\\ 0 & 2\end{bmatrix}$$

$$C_{11}=2, C_{12}=0, C_{21}=-(-4)=4, and\ C_n=1$$

$$C=\begin{vmatrix}2 & 0\\ 4 & 1\end{vmatrix}$$

$$adj \quad A=C^T$$

$$=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}$$

7. Ans: (c)

$AA^{-1} = I$

$$\begin{bmatrix}2 & -0.1\\ 0 & 3\end{bmatrix}\begin{bmatrix}\frac{1}{2} & a\\ 0 & b\end{bmatrix}=\begin{bmatrix}1 & 2a-0.1b\\ 0 & 3b\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$3b=1 Or \qquad b=\frac{1}{3}$$

$$2a-0.1b=0 \qquad or \qquad a=\frac{b}{20}$$

$$a+b=\frac{1}{3}+\frac{1}{3}\frac{1}{20}=\frac{7}{20}$$

8. Ans: (b) For Orthogonal Matrix, Transpose = Inverse of Matrix So,

$M^{-1} = M^T$

$$M^T=\begin{bmatrix}A & C\\ B & 0\end{bmatrix}=M^{-1}=\frac{1}{-BC}\begin{bmatrix}0 & -B\\ -C & A\end{bmatrix}$$

$$B=\frac{-C}{-BC} \qquad B=\frac{I}{B} or \qquad B=\pm 1$$

9. Ans: (b) Since $|kA| = k^n |A|$ where n is the order of the matrix

Then for A=5B and n=4

$|A| = |5B| = 5^4 |B| = 625 |B|$

Hence $\alpha = 625$

10. Ans: (c)

Here

$$A=\begin{bmatrix}1 & -4 & 7\\ 3 & 8 & -2\\ 7 & -8 & 26\end{bmatrix} \qquad B=\begin{bmatrix}14\\ 13\\ 5\end{bmatrix} \qquad [A:B]\begin{bmatrix}1 & -4 & 7 & :14\\ 3 & 8 & -2 & :13\\ 7 & -8 & 26 & :5\end{bmatrix}$$

the rank of [A] is 2 and rank of [A: B] is 3, hence the rank of Coefficient matrix A and Augmented matrix [A:B] is not same, so system of equation has no solution.

11. Ans: (a)

$$A=\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} \qquad A^2=\begin{bmatrix}3 & 3 & 3\\ 3 & 3 & 3\\ 3 & 3 & 3\end{bmatrix}, A^2=2A$$

Career Avenues

12. Ans: (d) Inverse of skew – symmetric, symmetric, Lower triangular, Upper triangular, diagonal matrix are same in nature. Option (d) is True

13. Ans: (c) Since the given set o eqaution is a homogenous equation and it is having non-zero solution so it has infinite solution and |A|=0

$$\begin{vmatrix} 3 & -1 & 1 \\ 15 & -6 & 5 \\ \lambda & -2 & 2 \end{vmatrix} = 0$$

$$-1(6-\lambda) = 0$$

$$\lambda = 6$$

14. Ans: (c) Here |A|=0, as the system of equation is Inconsistent
Then

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$\lambda = 3$$

15. Ans: (c) Since Addition of eigen values is equal to the sum of diagonal elements and product of eigen values is equal to the determinant of the matrix, so by reverse check 0 and 5 are the eigen values of given matrix.

16. Ans: (b) Since Addition of eigen values is equal to the sum of diagonal elements and product of eigen values is equal to the determinant o the matrix, so by reverse check 0,1 and 1 are the eigen values of given matrix.
OR by using the characteristics equation |A-λI|=0

$$\begin{vmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{vmatrix} = 0$$

$$(\lambda-1)\left[(4-\lambda)(-3-\lambda)-(2)(-6)\right] = 0$$

$$(\lambda-1)\left(-\lambda^2+\lambda\right) = 0$$

$$\lambda(\lambda-1)^2 = 0$$

$$\lambda = 0.1,1$$

17. Ans: (b) Since Addition of eigen values is equal to the sum of diagonal elements and product of eigen values is equal to the determinant of the matrix, so by reverse check 3, 3 and -5 are the eigen values of given matrix.

18. Ans: (c) The product of eigen values is equal to determinant of matrix.
1x3x4=12

19. Ans: (a) If the eigen value of a Matrix is x,y,z then the eigen value of inverse of will be 1/x, 1/y, 1/z .

20. Ans: (b) If the eigen value of a Matrix [A] is x,y,z then the eigen value of Matrix [2A]of will be 2x, 2y, 2z.

Career Avenues

21. Ans: (b)
If 5 and 1 are eigen value of S, then 5^2 and 1^2 will be eigen value of S^2 eigen value

22. Ans: (c) The characteristics equation is given by $|A-\lambda I|=0$

$$\begin{bmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix} = 0$$

$(-4-\lambda)(3-\lambda)-8=0$ Corresponding to λ = -5, we have

$\lambda = -5, 4$

$[A-(-5)I]X = 0$

$\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$ We get only one independent equation

$x_1 + 2x_2 = 0$

$\frac{x_1}{2} = \frac{x_2}{-1}$ giving eigenvector $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

23. Ans: (d)

Eigenvalues of A are	1,	-1,	0
Then eigenvalues of A^{100} are	1^{100},	$(-1)^{100}$,	$(0)^{100}$
i.e.,	1,	+1,	0
and eigenvalues of $I + A^{100}$ are	1+1	1+1	1+0
i.e	2,	2,	1
or	$\|I + A^{100}\|$ = 2 x 2 x 1 = 4		

24. Ans: (a) This question is based Caley- Hamilton theorem, matrix [P] will satisfy its characteristics equation $|P-\lambda I|=0$, Then
Then characteristic equation is

$$(\lambda-\lambda_1)(\lambda-\lambda_2)(\lambda-\lambda_3)=0$$

Or $(\lambda-1)(\lambda+2)(\lambda-3)=0$

Or $\lambda^3-2\lambda^2-5\lambda+6=0$

According to Cayley-Hamilton theorem

$$P^3-2P^2-5P+6I=0$$

Or $5P+2P^2-P^3=6I$

Or $P^{-1}=\frac{1}{6}(5I+2P-P^2)$

25. Ans: (b) As 1 and 3 are eigen values o Matrix [A] and roots of its characteristics equation $|A-\lambda I|=0$ then

$(\lambda-1)(\lambda-3)=0$

$\lambda^2-4\lambda+3=0$

So By Caley-Hamilton theorem λ=A then

$A^2-4A+3I=0$

$A^2=4A-3I$

$A^3=4A^2-3A$

$=4(4A-3I)-3A$

$A^3=13A-12I$

Career Avenues

SET-2

1. For Matrix [A] given below, The characterestics equation of Matrix is defined by

$$A=\begin{bmatrix}-3 & 2\\ -1 & 0\end{bmatrix}$$

a) (A+I) (A+2I) b) (2A-I) (A+I) c) (A-2I) (A+2I) d) (3A+I) (A-I)

2. For Matrix [A] given below, The value of A^9 will be equals to

$$A=\begin{bmatrix}-3 & 2\\ -1 & 0\end{bmatrix}$$

a) 300A + 104 I b) 154 A + 155 I c) e^{9A} d) 511A + 510I

3. If A is skew–symmetric matrix then which of following is not true ?

a) ρ (A) = 1 b) ρ (A) = 0 c) ρ (A) = 2 d) ρ (A) = 3

4. If $A = (a_{ij})_{7\times 7}$ is $\forall\ a_{ij} = -a_{ji}$

Then A^{-1} = ?

a) A^T itself b) Adj (A) itself c) $|A|^5A$ d) Non–existent

5. Which of following is true?

(a) Inverse of skew – symmetric matrix is symmetric

(b) Inverse of lower – triangular is upper triangular but vice-versa is not possible

(c) Inverse of upper – triangular is lower – triangular but vice-versa is not possible

(d) Inverse of diagonal matrix is always diagonal

6. If $A_{n\times n}$ is a square matrix then |Adj (Adj A)| = ?

a) $|A|^{2(n-1)}$ b) $|A|^n$ c) $|A|^{n-1}$ d) $|A|^{(n-1)}$

7. If $A_{n\times n}$ is a matrix such that $A^2 = I$ and set of equation can be written in the form of Ax = B then it has

a) No solution b) unique solution

c) finite independent solution d) infinite solution

8. The eigen vector of matrix A= $\begin{bmatrix}2 & 1\\ 0 & 1\end{bmatrix}$ are in form $\begin{bmatrix}1\\ x\end{bmatrix}, \begin{bmatrix}1\\ Y\end{bmatrix}$ Then x + y = ?

a) 0 b) 1 c) −1 d) 2

Career Avenues

9. If $A_{3\times3}$ has $|A| = 9$ then what is $|\det| (3A)^{-1}|$

a) $\frac{1}{27}$ b) 27 c) $\frac{1}{81}$ d) $\frac{1}{245}$

10. If $A_{n\times n} = (a_{ij})$ is upper triangular matrix then $|A|$ is

a) $\prod_{K=1}^{n}(-1)a_{kk}$ b) $\prod_{K=1}^{n}a_{kk}$ c) $\sum_{K=1}^{n}(-1)a_{kk}$ d) $\sum_{K=1}^{n}a_{kk}$

11. Rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix}$ is

a) 0 b) 1 c) 2 d) 3

12. Rank of matrix $A_2 \begin{bmatrix} a & a & a \\ a & a & 0 \\ a & a & a \end{bmatrix}$ Where $a \neq 0$, is

a) 0 b) 1 c) 2 d) 3

13. If points (x, y,), (x_2, y_2) and (x_3, y_3) lies on same line, then rank of $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{bmatrix}$

a) 0 b) 1 c) less than 3 d) 3

14. If A is n × n matrix and $|A| \neq 0$, adj (A) = ?

a) A^{-1} b)$|A|. A^{-1}$ c) $\frac{A^{-1}}{|A|}$ d) A

15. If $A_{3 \times 3}$ matrix has |A| = 8 then |adj (A)| = ?

a) 8 b) $\sqrt{8}$ c) 64 d) 3/2

16. Which of the following is false?

a) A square matrix A is symmetric, if A' = A.

b) A square matrix A is skew-symmetric, if A' = - A.

c) The matrix $\begin{bmatrix} 6 & 9 \\ -4 & -6 \end{bmatrix}$ is nilpotent.

d) The matrix $\begin{bmatrix} i & 1-i \\ -1-i & -i \end{bmatrix}$ is Hermitian.

17. The matrix $\begin{bmatrix} 0 & -4 & 1 \\ 4 & 0 & -5 \\ -1 & 5 & 0 \end{bmatrix}$ is

a) Orthogonal b) idempotent

c) skew-symmetric d) symmetric

Career Avenues

18. The matrix $A = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$ is

a) orthogonal

b) unitary

c) idempotent

d) skew-symmetric

19. The number of values of k for which the system of equations $(k + l)x + 8y = 4k$; $kx + (k + 3)y = 3k - 1$ has infinitely many solutions is

a) 0

b) 1

c) 2

d) Infinite

20. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \left[\frac{1}{6}\left(A^2 + cA + dI\right)\right]$, the value of c and d are

a) (-6, -11)

b) (6, 11)

c) (-6, 11)

d) (6, -11)

21. The matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is

a) involutory

b) orthogonal

c) nilpotent of index 2

d) idempotent

22. Which of the following if false?

a) If A is a nilpotent matrix, then $A^n = 0$.

b) If A is an orthogonal matrix, the A^{-1} is also orthogonal.

c) If A an unitary matrix, then A^{-1} is not unitary.

d) If A and B are two unitary matrices, then the product of A and is also unitary.

Career Avenues

23. If $A = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{bmatrix}$, then which of the following is false?

a) ρ (A) = 1 if A = b = c

b) ρ (A) = 2 if A = b $\neq$ c

c) ρ (A) = 3 if a $\neq$ b $\neq$ c

d) ρ (A) is 3 always

24. If there is a non-zero minor of order (r) of the matrix A, and all the minors of higher orders are zero, then

a) ρ (A) = r

b) ρ (A) $\neq$ r

c) ρ (A) < r

d) ρ (A) > r

25. If 3x + 2y + z = 0;x + 4y + z = 0 ; 2x + y + 4z = 0 be a system of equations, then

a) It is inconsistent.

b) It has only the trival solution x = 0, y = 0, z = 0.

c) It can be reduced to a single equation and so a solution does not exist.

d) The determinant of the matrix of coefficient is zero.

Career Avenues

Solutions (SET-2)

1. Ans: (a) its characteristics equation $|A-\lambda I|=0$ is given by

$$\begin{bmatrix} -3-\lambda & 2 \\ -1 & 0-\lambda \end{bmatrix} = 0$$

$$(-3-\lambda)(-\lambda)+2=0$$

$$(\lambda+1)(\lambda+2)=0$$

Then by caley-Hamilton theorem $\lambda=A$ and
$(A+I)\ (A+2I)$

2. Ans: (d) By Caley –Hamilton theorem
3. Ans: (a)

$\rho(A) = 1$ as $A_{2\times 2} \neq 0$

but in case of $A_{3\times 3}$, $|A| = 0$

4. Ans: (d)

For skew symmetric matrix of odd order, it's determinant $|A| = 0$

A^{-1} doesn't exist

5. Ans: (d)

Inverse of skew – symmetric, symmetric, Lower triangular, Upper triangular, diagonal matrix are same in nature. Option (d) is True

6. Ans: (d)

$|\text{Adj (adj A)}| = |\text{Adj (B)}| = |B|^{n-1}$

But $|B| = |A|^{n-1} \Rightarrow |\text{adj (adj A)}| = |A|^{(n-1)}$

7. Ans: (b)

$A^2 = A.\ A. = I \Rightarrow |A|.\ |A| = I \Rightarrow |A|^2 = 1$

$\Rightarrow |A| = \pm 1 \neq 0$

$\therefore\ \rho(A) = \rho(A/B) = n$

$\therefore$ Unique solution

8. Ans: (c)

In upper triangular matrix

$\propto = 2, 1$

$|A - \lambda I = 0$

$$\text{At } \lambda = 1 \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_1 = \begin{bmatrix} K_1 \\ -1 \end{bmatrix} - (1)$$

$$\lambda = 2 \rightarrow \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_2 = \begin{bmatrix} K_2 \\ 0 \end{bmatrix} - (2)$$

$$\begin{bmatrix} 1 \\ x \end{bmatrix} \leftrightarrow \begin{bmatrix} K_1 \\ -K_1 \end{bmatrix} \Rightarrow x = -1$$

Career Avenues

$$\begin{bmatrix}1\\Y\end{bmatrix}\leftrightarrow\begin{bmatrix}K_2\\0\end{bmatrix}\Rightarrow Y=0$$

$\therefore$ x + y = -1+0= -1

9. Ans: (d)

|A| = 0

Using property $|KA| = K^n |A|$

$$\Rightarrow \quad |3A|=3^3|A|=27\times 9=243$$

$$\therefore\left((3A)^{-1}\right)=\frac{1}{243}$$

10. Ans: (b)

In triangular (upper or lower) matrix, determinant is product of principle diagonal element.

11. Ans: (b) $R_3 - (R_2 + R_1) \rightarrow A = \begin{bmatrix}1 & 2 & 3\\4 & 8 & 12\\0 & 0 & 0\end{bmatrix}$

$R_2 - 4R_1 \rightarrow A = \begin{bmatrix}1 & 2 & 3\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$ $\therefore$ Rank = 1 Hence [b]

12. Ans: (c) $R_1 = R_3 \Rightarrow |A| = 0$

$\therefore$ Rank = 2

13. Ans: (c)

As all point lies on same line, y = mx

$\therefore R_2 = mR_1 \Rightarrow |A| = 0$

$\therefore$ Rank < 3

14. Ans: (b) A adj (A) = $|A|. I_n$

$A^{-1}. A. adj (A) = A^{-1}| A|. I_n = |A| . A^{-1}$

Adj (A) = $|A|. A^{-1}$

15. Ans: (c)

Using property $|Adj (A)| = |A|^{n-1}$

$|adj (A)| = |A|^{3-1} = 8^2 = 64$

16. Ans: (d) Since in any Hermitian matrix, the diagonal elements must be purely real, hence $\begin{bmatrix}i & 1-i\\-1-i & -i\end{bmatrix}$ cannot be Hermitian, hence (d) is false statement.

17. Ans: (c) Let $A = \begin{bmatrix}0 & -4 & 1\\4 & 0 & -5\\-1 & 5 & 0\end{bmatrix}$

Career Avenues

$$A' = \begin{bmatrix} 0 & 4 & -1 \\ -4 & 0 & 5 \\ 1 & -5 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 4 & 1 \\ 4 & 0 & -5 \\ -1 & 5 & 0 \end{bmatrix} = - A$$

$\therefore$ A is a skew-symmetric matrix.

18. Ans: (a) From question (14), $AA' = I$

$\therefore$ A is an orthogonal matrix.

19. Ans: (b) For infinitely many solutions, the two equations become identical

$$\Rightarrow \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1} \Rightarrow k = 1$$

20. Ans: (c) We have, $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$

$\Rightarrow 6AA^{-1} = A^3 + cA^2 + Dai$

$\Rightarrow A^3 + cA^2 + dA = 6I = 0$

We find that, $A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -11 & 19 \\ 0 & -38 & 46 \end{bmatrix}$$

$\therefore A^3 + cA^2 + dA - 6I = 0$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -11 & 19 \\ 0 & -38 & 46 \end{bmatrix} + c \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} + d \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} - 6 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 + c + d - 6 & 0 & 0 \\ 0 & -11 - c + d - 6 & 19 + 5c + d \\ 0 & -38 - 10c - 2d & 46 + 14c + 4d - 6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow \quad 1 + c + d - 6 = 0$

$= 11 - c + d - 6 = 0$

Career Avenues

$\Rightarrow$ $c + d = 5$

$- c + d = 17$

On solving, we get, $c = - 6, d = 11$

$$\Rightarrow \quad \frac{1}{6}\begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{6}\left\{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} + \begin{bmatrix} c & 0 & 0 \\ 0 & c & c \\ 0 & -2c & 4c \end{bmatrix} + \begin{bmatrix} d & 0 & 0 \\ 0 & d & 5 \\ 0 & 0 & d \end{bmatrix}\right\}$$

$$\Rightarrow \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 + c + d & 0 & 0 \\ 0 & -1 + c + d & 5 + c \\ 0 & -10 - 2c & 14 + 4c + d \end{bmatrix}$$

$\Rightarrow$ $1 + c + d = 6$

and $-1 = 5 + c$

$\Rightarrow$ $c = - 6$ and $d = 11$

21. Ans: (d) Since $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

$$\therefore A.A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} x \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

$\therefore A^2 = A$

Hence A is an idempotent matrix.

22. Ans: (c)

Since (a), (b) and (d) are well known properties. We will check (c).

Let A be unitary, then $AA^* = A^*A = 1$

Or $(AA^*)^{-1} = (A^*A)^{-1} = I^{-1} = 1$

Or $(A^*)^{-1} A^{-1} = (A^*A)^{-1} = I^{-1} = 1$

Or $(A^{-1})^* A^{-1} = A^{-1} (A^{-1})^* = 1$

$\therefore$ A^{-1} is unitary.

23. Ans: (d) $|A| = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b - a & c - a \\ a^2 & b^2 - a^2 & c^2 - a^2 \end{vmatrix}$, $\begin{matrix} C_2 \to C_2 - C_1 \\ C_3 \to C_3 - C_1 \end{matrix}$

Career Avenues

$$= (b - a)(c - a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2 & b + a & c + a \end{vmatrix}$$

$= (b - a)(c - a)(c - b)$

$\therefore$ Case I when a = b = c, then $|A| = 0$

$\therefore \rho(A) \leq 2$

Also all minors of order 2 are equal to zero.

$\therefore \rho(A) \leq 1$ ….(i)

But $|1| = 1 \neq 0$, all minor of order one are not zero.

$\therefore \rho(A) \leq 1$ ….. (ii)

From equations (i) and (ii), $\rho(A) = 1$, if a = b = c.

Hence (a) is correct.

Case II Two out of a, b, c are equal but different from third.

$\therefore \quad |A| = (A - b)(A - c)(c - b) = 0$

$\therefore \quad \rho(A) \leq 2$ ….(iii)

Suppose A = b ≠ c, then $\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} = c - b \neq 0$

$\therefore$ The minor $\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} \neq 0$ or order two

$\therefore \quad \rho(A) \leq 2$

$\therefore$ From equations (iii) and (iv)

$\rho(A) = 2$

Hence (b) is correct.

Case III a, b, c are all distinct.

$|A| \neq 0$

$\rho(A) = 3$

Because $|A|$ has no minor of order greater than three.

Hence (c) is correct.

Thus the correct answer of this question is (d)

Note For examination purpose

$|A| = (a - b)(a - c)(c - b)$.

Now we cannot say that $|A| = 0$ always, hence option (d) should be ticked, and no need to check (a), (b) and (c).

Career Avenues

24. Ans: (a)

25. Ans: (b) Given equations are

$3x + 2y + z = 0, \quad x + 4y + z = 0, \quad 2x + y + 4z = 0$

Here, $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $o = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\therefore \quad |A| = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix} = 34 \neq 0$

α. ρ (A) = 3

β. ρ (A) is equal to the number of variables. Hence, there is only trivial solution x = 0, y = 0, z = 0.

Career Avenues

SET-3

1. A is a 3×4 real matrix and Ax= b is an inconsistent system of equations. The highest possible rank of A is

   a)1 b)2 c)3 d)4

2. The matrix is $\begin{bmatrix} \frac{1}{2}(1+i) & -\frac{1}{2}(1-i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix}$ is

   a) orthogonal b) unitary

   c) Skew-Hermitian d) None of these

3. If A and B are any two square symmetric then ..... is a skew-symmetric matrix?

   a) $\frac{A + A'}{2}$ b) $\frac{A' + B'}{2}$

   c) $\frac{A' - B'}{2}$ d) $\frac{B - B'}{2}$

4. If there exist a non-zero minor of order (r), then rank of A is

   a) equal to r

   b) greater or equal to r

   c) less than r

   d) less than or equal to r

5. If the number of variables in the non-homogeneous system AX = B is n, then the system possesses a unique solution, if

   a) ρ (A) < ρ [A: B]

   b) ρ (A) = ρ [A: B] = n

   c) ρ (A) > ρ [A: B]

   d) None of these

6. If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then

   a) $|\text{Adj } A| = |A|$ b) $|\text{Adj } A| = |A^{-1}|$

   c) $|\text{Adj } A^{-1}| = |A|$ d) $|\text{Adj } A^{-1}| = (|A|)^{-1}$

Career Avenues

7. Consider the matrices $X_{(4\times3)}, Y_{(4\times3)}$ *and* $P_{(2\times3)}$. The order or $[P(X^TY)^{-1}P^T]^T$ will be

a) (2×2) b) (3×3) c) (4×3) d) (3×4)

8. A square matrix B is skew-Symmetric if

a) $B^T=-B$ b) $B^T=B$ c) $B^{-1}=B$ d) $B^{-1}=B^T$

9. Let, $A=\begin{bmatrix}2 & -0.1\\ 0 & 3\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}\frac{1}{2} & a\\ 0 & b\end{bmatrix}$ then (a + b) =

a) $\frac{7}{20}$ b) $\frac{3}{20}$ c) $\frac{19}{60}$ d) $\frac{11}{20}$

10. Given the matrix $\begin{bmatrix}-4 & 2\\ 4 & 3\end{bmatrix}$, the eigen vector is

a) $\begin{bmatrix}3\\2\end{bmatrix}$ b) $\begin{bmatrix}4\\3\end{bmatrix}$ c) $\begin{bmatrix}2\\-1\end{bmatrix}$ d) $\begin{bmatrix}-2\\1\end{bmatrix}$

11. The eigen values and the corresponding eigen vectors of a 2×2 matrix are given by

Eigen values	Eigenvector
$\lambda_1=8$	$v_1=\begin{bmatrix}1\\1\end{bmatrix}$
$\lambda_2=4$	$v_1=\begin{bmatrix}1\\-1\end{bmatrix}$

The matrix is

a) $\begin{bmatrix}6 & 2\\ 2 & 6\end{bmatrix}$ b) $\begin{bmatrix}4 & 6\\ 6 & 4\end{bmatrix}$ c) $\begin{bmatrix}2 & 4\\ 4 & 2\end{bmatrix}$ d) $\begin{bmatrix}4 & 8\\ 8 & 4\end{bmatrix}$

12. Consider the matrix as given below:

$$\begin{bmatrix}1 & 2 & 3\\ 0 & 4 & 7\\ 0 & 0 & 3\end{bmatrix}$$

Which one of the following options provides the CORRECT values of the eigen values of the matrix?

a) 1, 4, 3 b) 3, 7, 3 c) 7, 3, 2 d) 1, 2, 3

Career Avenues

13. Match the item in columns I and II.

Column I	Column II
P. Singular matrix	1. Determinant is not defined
Q. Non-square matrix	2. Determinant is always one
R. Real symmetric	3. Determinant is zero
S. Orthogonal matrix	4. Eigen values are always real
	5. Eigen values are not defined

a) P-3, Q-1, R-4, S-2 b) P-2, Q-3, R-4, S-1
c) P-3, Q-2, R-5, S-4 d) P-3, Q-4, R-2, S-1

14. Find the trace of

$$[A]=\begin{bmatrix}15 & 6 & 7\\ 2 & -4 & 2\\ 3 & 2 & 6\end{bmatrix}$$

15. The sales of jeans are given by make (rows) and quarters (columns) for ABC Departmental Store location A, as shown below.

$$[A]=\begin{bmatrix}25 & 20 & 4 & 2\\ 5 & 10 & 15 & 25\\ 6 & 16 & 7 & 27\end{bmatrix}$$

where the rows represent the sale of Levis, Newport and Pepe jeans, and the columns represent the quarter number 1, 2, 3, 4.
Find the total yearly revenue of store A if the prices of jeans vary by quarters as follows.

$$[B]=\begin{bmatrix}33.25 & 20.01 & 35.02 & 30.05\\ 40.19 & 38.02 & 41.03 & 38.23\\ 25.03 & 22.02 & 27.03 & 22.95\end{bmatrix}$$

where the rows represent the cost of each jeans made by Levis, Newport and Pepe, and the columns represent the quarter numbers.

Career Avenues

Solution (Set-3)

1. Ans: (b) highest possible rank of A=2, as Ax=b is an inconsistent system.

2. Ans: (b) Let $A = \begin{bmatrix} \frac{1}{2}(1=i) & -\frac{1}{2}(1-i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1+i) \end{bmatrix}$

$\therefore A^* = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ -\frac{1}{2}(1+i) & \frac{1}{2}(1+i) \end{bmatrix}$

$\therefore AA^* = \begin{bmatrix} \frac{1}{2}(1+i) & -\frac{1}{2}(1-i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix} x \begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \\ -\frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

$\therefore AA^* = 1$

$\therefore$ A is an unitary matrix.

3. Ans: (d) $\frac{1}{2}(B - B') = \frac{1}{2}(B' - B) = -\frac{1}{2}(B - B')$

$\therefore \frac{1}{2}(B - B')$ is skew-symmetric

4. Ans: (a)
5. Ans: (b)
6. Ans: (a) If A is n $\times$ n order matrix and $|A| \neq 0$, then $|\text{adj } A| = |A|^{n-1}$ = Here, n = 2, hence $|\text{adj } A| = |A|$
7. (a) $[P_{2\times3}(X^T{}_{3\times4}Y_{4\times3})^{-1}P^T{}_{3\times2}]^T$

$=[P_{2\times3}Z^{-1}{}_{3\times3}P^T{}_{3\times2}]^T$ Take Z=XY

$=[T_{2\times2}]^T = [T']_{2\times2}$ $\begin{bmatrix} T = PZ^{-1}P^T \\ T' = T^T \end{bmatrix}$

8. (a) $B^T=-B$
9. (a) we know $AA^{-1}=I_2$

$\Rightarrow \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix} = \begin{bmatrix} 1 & 2a-0.1b \\ 0 & 3b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

b = 1/3 and a=1/60

a + b = 7/20

Career Avenues

10. Ans: (c)
Characteristic equation
$|A-\lambda I_2|=0$
$\Rightarrow \begin{vmatrix} -4-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix}=0$
$\Rightarrow \lambda=-5,4$
Take λ =-5, then AX= λ X becomes
$\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} -5x_1 \\ -5x_2 \end{bmatrix}$
$\Rightarrow \begin{bmatrix} -4x_1 & 2x_2 \\ 4x_1 & 3x_2 \end{bmatrix}=\begin{bmatrix} -5x_1 \\ -5x_2 \end{bmatrix}$
$\therefore \left.\begin{matrix} -4x_1+2x_2=-5x_1 \\ -4x_1+3x_2=-5x_2 \end{matrix}\right\} \Rightarrow x_1=-2x_2$
$\therefore$ if x_2=-1 then x_1=2
$\therefore \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ is eigen vector corresponding to λ = -5

11. Ans: (a) We know sum of eigen values =trace(A)=Sum of diagonal elements of A.
Therefore $\lambda_1+\lambda_2$ =8+4=12
Option (a) gives, trace (A)=6+6=12.

12. Ans: (a) It's an upper triangular matrix and sum o diagonal element is equals to um o eigen values.

13. (a) (P) Singular matrix →Determinant is zero
(Q) Non-square matrix →Determinant is not defined
(R) Real symmetric → Eigen values are always real
(S) Orthogonal → Determinant is always one

14. Solution
$tr[A]=\sum_{i=1}^{3} a_{ii}$
= (15) + (-4) + (6)
= 17

15. To find the total jeans sales of store A for the whole year, we need to find the sales of each brand of jeans for the whole year and then add to find the total sales. To do so, we need to rewrite the price matrix so that the quarters are in rows and the brand names are in the columns, that is, find the transpose of [B].
$[C]=[B]^T$
$=\begin{bmatrix} 33.25 & 30.01 & 35.02 & 30.05 \\ 40.19 & 38.02 & 41.03 & 38.23 \\ 25.03 & 22.02 & 27.03 & 22.95 \end{bmatrix}^T$
$[C]=\begin{bmatrix} 33.25 & 40.19 & 25.03 \\ 30.01 & 38.02 & 22.02 \\ 35.02 & 41.03 & 27.03 \\ 30.05 & 38.23 & 22.95 \end{bmatrix}$

Career Avenues

Recognize now that if we find[A][C] , we get
[D] = [A] [C]

$$= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \begin{bmatrix} 33.25 & 40.19 & 25.03 \\ 30.01 & 38.02 & 22.02 \\ 35.02 & 41.03 & 27.03 \\ 30.05 & 38.23 & 22.95 \end{bmatrix}$$

$$= \begin{bmatrix} 1597 & 1965 & 1193 \\ 1743 & 2152 & 1325 \\ 1736 & 2169 & 1311 \end{bmatrix}$$

The diagonal elements give the sales of each brand of jeans for the whole year, that is

$d_{11} = \$1597$ (Levis sales)

$d_{22} = \$2152$ (Newport sales)

$d_{33} = \$1311$ (Cooper sales)

The total yearly sales of all three brands of jeans are

$$\sum_{i=1}^{3} d_{ii} = 1597 + 2152 + 1311$$

$$= \$5060$$

and this is the trace of the matrix [A].

Career Avenues

Previous Year Gate Solved Paper Linear Algebra

1. Consider Matrix $A=\begin{bmatrix} k & 2k \\ k^2-k & k^2 \end{bmatrix}$ and vector $x=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the number of distinct real values of k for which the equation AX=0 has infinitely many solution is… **Gate-18 (1-Mark)**

2. The Rank of the Matrix **Gate-17 (1-Mark)**

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

is ……..

3. The Rank of the Matrix is **Gate-17 (1-Mark)**

$$M=\begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$$

a) 0 b) 1 c) 2 d) 3

4. Consider a 5x5 Matrix **Gate-17 (1-Mark)**

$$A=\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}$$

It is given that A has only one real eigen value Then the real eigen value of A is

a) -2.5 b) 0 c)15 d) 25

5. Consider a 2x2 square matrix **Gate-16 (1-Mark)**

$$A=\begin{bmatrix} \sigma & x \\ \omega & \sigma \end{bmatrix}$$

Where x is known. If the eigen values of the matrix A are $(\sigma+j\omega)$ and $(\sigma-j\omega)$

a) $+j\omega$ b) $-j\omega$ c) $+\omega$ d) $-\omega$

6. The Matrix A has det(A)=100 **Gate-16 (2-Mark)**

$$A=\begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$$

And trace (A) = 14. The value of |a-b| is ……..

Career Avenues

7. The value of x for which the matrix **Gate-16 (1-Mark)**

$$A=\begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9+x \end{bmatrix}$$

has zero as an eigen value is ……

8. A sequence x[n] is specified as **Gate-16 (2-Mark)**

$$\begin{bmatrix} x(n) \\ x(n-1) \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \text{ for } n \geq 2.$$

The initial conditions are x [0] = 1, x [1] = 1, and x[n]=0 for n<0. The value of x [12] is ……..

9. Let $M^4 = I$ (Where **I** denote the identity matrix) and $M \neq I, M^2 \neq I$ and $M^3 \neq I$ then or any natural number k, M^{-1} equals to **Gate-16 (1-Mark)**

a) M^{4k+1}
b) M^{4k+2}
c) M^{4k+3}
d) M^{4k}

10. The determinant of $A^T A^{-1}$is ……, **Gate-15 (1-Mark)**

Where $A=\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

a) $\sec^2 x$
b) $\cos^4 x$
c) 1
d) 0

11. The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$ **Gate-15 (1-Mark)**

a) 5+j b) 5-j c) 1-5j d) 1+5j

12. The value of p such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigenvector o the matrix

$$\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$$ is **Gate-15 (1-Mark)**

13. Consider a system of linear equations **Gate-15 (1-Mark)**

x - 2y +3z= -1
x -3y+4z =1 and
-2x+ 4y-6z=k
The value of k for which the system has infinitely many solution is…….

Career Avenues

14. Which of the following statements is NOT true or a square Matrix A? **Gate-14 (2-Mark)**

a) If A is upper triangular, the eigen values of A are the diagonal elements of it
b) If A is real symmetric, the eigen values of A are always real and positive.
c) If A is real, the eigen values of A and Transpose of A are always the same.
d) If all the principal minors of A are positive, all the eigen values of A are also positive.

15. The maximum value of determinant among all 2x2 real symmetric matrices with traces 14 is ……. **Gate-14 (2-Mark)**

16. The system of linear equations **Gate-14 (2-Mark)**

$$\begin{bmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 14 \end{bmatrix}$$ has

a) A unique solution
b) Ininetly many solutions
c) No solution
d) Exactly two solution

17. The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is…… **Gate-14 (1-Mark)**

18. A real (4x4) matrix A satisfies the equation $A^2 = I$ where I is the (4x4) identity matrix. The positive eigen value of A is…… **Gate-14 (1-Mark)**

19. The minmum eigen value of the following matrix is **Gate-13 (1-Mark)**

$$\begin{bmatrix} 3 & 5 & 2 \\ 5 & 12 & 7 \\ 2 & 7 & 5 \end{bmatrix}$$

a) 0 b) 1 c) 2 d) 3

20. Let A be an m x n matrix and B an n x m matrix. It is given that determinant $(I_m + AB)$ = determinant $(I_n + BA)$ where I_k is the k x k identity matrix.

Using the above property, the determinant of the matrix given below is

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$ **Gate-13 (2-Mark)**

a) 2 b) 5 c) 8 d) 16

21. Given that **Gate-12 (2-Mark)**

$A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ the value of A^3 is

a) 15 A + 12 I b) 19 A + 30 I c) 19 A + 30 I d) 17 A + 21 I

Career Avenues

22. The system of equations **Gate-11 (2-Mark)**

$x + y + z = 6$

$x + 4y + 6z = 20$

$x + 4y + \lambda z = \mu$

has no solution for values of λ and μ given by

a) $\lambda = 6, \mu = 20$

b) $\lambda = 6, \mu \neq 20$

c) $\lambda \neq 6, \mu = 20$

d) $\lambda \neq 6, \mu \neq 20$

23. The eigen value of a skew-symmetric matrix are **Gate-10 (1-Mark)**

a) Always zero

b) Always pure imaginary

c) Either zero or pure imaginary

d) Always real

24. The eigen values of the following matrix are **Gate-09 (2-Mark)**

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

a) 3, 3 + 5j, 6 - j

b) -6 + 5j, 3 + j, 3 - j

c) 3 + j, 2 – j, 5 + j

d) 3, -1 + 3 j, -1-3j

25. The system of linear equations **Gate-08 (1-Mark)**

4x+2y=7

2x+ y=6 has

a) A unique solution

b) No solution

c) An infinite number of solutions

d) Exactly two distinct solutions

26. All the four entries of the 2x2 matrix **Gate-08 (1-Mark)**

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$$

are non-zero, and one of its Eigen value is zero. Which of the following statement is true?

a) $P_{11}P_{22} - P_{12}P_{21} = 1$

b) $P_{11}P_{22} - P_{12}P_{21} = -1$

c) $P_{11}P_{22} - P_{12}P_{21} = 0$

d) $P_{11}P_{22} + P_{12}P_{21} = 0$

27. For the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ the eigen value corresponding to the eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ is **Gate-06 (2-Mark)**

a) 2 b) 4 c) 6 d) 8

Career Avenues

Solutions

1. AX=0 has infinitely many solutions, So |A|=0

$$\begin{bmatrix} k & 2k \\ k^2-k & k^2 \end{bmatrix} \; k^3-2k^3+2k^2=0$$

$$k^2(2-k)=0$$

K = 0,2 $\Rightarrow$ "two" distinct values of k

2.

$$A=\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \; R_4 \to R_4 + R_1 = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \quad R_4 \to R_4 + R_3$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \; R_4 \to R_4 + R_2 = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \; R_2 \leftrightarrow R_3 \; and \; R_5 \to R_5 + R_4$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence rank = 4

3.

$$M=\begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}_{R_1 \leftrightarrow R_2:} \begin{bmatrix} 1 & 0 & 2 \\ 5 & 10 & 10 \\ 3 & 6 & 6 \end{bmatrix}_{R_2 \leftarrow R_2 - 5R_1 \; and \; R_3 \leftarrow R_3 - 3R_1:}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 10 & 0 \\ 0 & 6 & 0 \end{bmatrix}_{R_3 \leftarrow R_3 - \frac{6}{10}R_2:} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Which is in Echelon from
Rank of Matrix M is,
$\rho(M)=2$

4.

$$|A-\lambda I|=0$$

$$= \begin{bmatrix} 1-\lambda & 2 & 3 & 4 & 5 \\ 5 & 1-\lambda & 2 & 3 & 4 \\ 4 & 5 & 1-\lambda & 2 & 3 \\ 3 & 4 & 5 & 1-\lambda & 2 \\ 2 & 3 & 4 & 5 & 1-\lambda \end{bmatrix}$$

Sum of all elements in any one row must be zero
i. e 15-λ=0

Career Avenues

λ=15

5. Trace = Sum of eigen values

$A=\begin{bmatrix}\sigma & x\\ \omega & \sigma\end{bmatrix}$ $\qquad 2\sigma=\sigma+j\omega+\sigma-j\omega$

And Product of eigen values = determinant of Matrix

$\sigma^2-x\omega=(\sigma+j\omega)(\sigma-j\omega)=\sigma^2+\omega^2$

Then $x=-\omega$

6. Trace of A=14

a+5+2+b=14

a+b=7 ……..(i)

det(A)=100

5x2xaxb=100

10ab=100

ab=10 ……(ii)

from equation (i) and (ii)

a=2 and b=5

then |a-b|=|2-5|= 3

7. A has an eigen value equals to zero therefore |A|=0

$\begin{bmatrix}3 & 2 & 4\\ 9 & 7 & 13\\ -6 & -4 & -9+x\end{bmatrix}=0$

then x = 1

8. For $A=\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}$ characteristics equation |A-ʎI|=0

$\begin{bmatrix}1-\lambda & 1\\ 1 & -\lambda\end{bmatrix}=0$ $\quad -\lambda+\lambda^2-1=0$

$\lambda^2-\lambda-1=0$

By Caley Hamilton theorem ʎ=A

$A^2=A+I$

$A^4=A^2+2A+I$

$=A+I+2A+I=2A+2I$

$A^2 - A - I = 0$ $\quad A^8=9A^2+12A+4I \quad =\begin{bmatrix}233 & 144\\ 144 & 89\end{bmatrix}$

$=9(A+I)+12A+4I$

$=21A+13I$

$A^{12}=A^4\cdot A^8=144A+89I$

$\begin{bmatrix}x[12]\\ x[11]\end{bmatrix}=\begin{bmatrix}233 & 144\\ 144 & 89\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix}$

$x[12]=233$

Career Avenues

9. c) Given that $M^4 = I$, $M^{4k} = I\ M^{4\,(k+1)} = I$

$$M^{-1} \times I = M^{4\,(k-1)} \times M^{-1}$$
$$M^{-1} = M^{4k+3}$$

10. c) Since $|AB|=|A||B|$

$|A^T A^{-1}| = |A^T|\ |A^{-1}|$

$$= |A| \times \frac{1}{|A|} = 1$$

$$\left(Note: |A^T| = |A| \; and \; |A^{-1}| = \frac{1}{|A|} \right)$$

11. b) For a matrix containing complex number, eigen values are real if and only if

$$A = A^{\theta} = \left(\overline{A}\right)^T$$

$$A = \begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

$$A^{\theta} = \left(\overline{A}\right)^T = \begin{bmatrix} 10 & \overline{x} & 4 \\ 5-j & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$ by comparing these x=5-j

12. A X = ʎ X

$$\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 12 \\ p+7 \\ 36 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\frac{p+7}{12} = 2 \Rightarrow p = 17$$

13.

$$x - 2y + 3z = -1$$
$$x - 3y + 4z = 1, and$$
$$-2x + 4y - 6z = k$$

$$[A:B] = \begin{bmatrix} 1 & -2 & 3 & : & -1 \\ 1 & -3 & 4 & : & 1 \\ -2 & 4 & 6 & : & k \end{bmatrix} \overset{R_2 \to R_2 - R_1 . R_3 \to R_3 + 2R_1}{\begin{bmatrix} 1 & -2 & 3 & : & -1 \\ 0 & -1 & 1 & : & 2 \\ 0 & 0 & 0 & : & k-2 \end{bmatrix}}$$

For infinite many solution

$$\rho(A:B) = \rho(A)$$
$$= r < number\ of\ variables$$
$$\rho(A:B) = 2$$
$$k - 2 = 0$$
$$k = 2$$

Career Avenues

14. b)

15.
$$A=\begin{bmatrix} a & x \\ x & b \end{bmatrix} \xrightarrow{Trans} \begin{bmatrix} a & x \\ x & b \end{bmatrix}$$
Let (a) and (b) are the elements of principal diagonal
$\Rightarrow |A| = ab - x^2$
$\frac{d|A|}{dx} = -2x = 0 \quad (x = 0)$
So, |A| = a. b = maximum 1 when (a) and (b) should be equal.
= (7 x 7)
= 49 (become 7 + 7 = 14)

16. b)
$\begin{bmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 14 \end{bmatrix}$ Augmented matric is $\begin{bmatrix} 2 & 1 & 3 & 5 \\ 3 & 0 & 1 & -4 \\ 1 & 2 & 5 & 14 \end{bmatrix}$
$2R_2 - 3R_1$. $2R_3 - R_1$
$\begin{bmatrix} 2 & 1 & 3 & 5 \\ 0 & -3 & -7 & -23 \\ 0 & 3 & 7 & 23 \end{bmatrix} R_3 + R_2 \begin{bmatrix} 2 & 1 & 3 & 5 \\ 0 & -3 & -7 & -23 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
Rank of A=2, Rank of Augmented matrix=2 and Number of variables=3
Therefore, the system possesses infinitely many solutions.

17. Determinant of A=5, Determinant of B=40,
therefore, Determinant of AB=|A| |B| =5x40=200

18. Since square of A is unity matrix then all the diagonal element of unity matrix will be equal to 1 so, sum of diagonal element is equal to sum of eigen values and product of eigen values will be equal to determinant of matrix. i.e.
$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1+1+1+1 = 4$
$\lambda_1 \lambda_2 \lambda_3 \lambda_4 = 1$
So λ=1 satisfy the equation.

19. a) The characteristics equation of the matrix is $|A-\lambda I| = 0$
$$\begin{bmatrix} 3-\lambda & 5 & 2 \\ 5 & 12-\lambda & 7 \\ 2 & 7 & 5-\lambda \end{bmatrix} = 0$$
$(3-\lambda)\left[(12-\lambda)(5-\lambda)-49\right]$
$-5\left[5(5-\lambda)-14\right]+2\left[35-2(12-\lambda)\right]=0$
$\lambda\left(\lambda^2-24\lambda+37\right)=0$

Career Avenues

$\lambda_1 = 0$

$\lambda_2, \lambda_3 = \dfrac{24 \pm \sqrt{(24)^2 - 4(37)}}{2}$

$\lambda_2 = 22$

$\lambda_3 = 2$

$\lambda_{mn} = 0$

20. b)

Take the determinant of given matrix

$$|P| = 2[2(4-1)-1(2-1)+1(1-2)] - 1[1(4-1)-1(2-1)+1(1-2)] + 1[1(2-1)-2(2-1)+1(1-1)] - 1[1(1-2)-2(1-2)+1(1-1)]$$

= 2 [6-1-1] -1 [3-1-1] + 1 [1-2+0] -1 [-1+2+0]

= 29(4) -1 (1) + 1 (-1) -1 (1) =8 -1-1-1

=5

21. b) The Characteristics equation of A will be |A-ʎI|=0

$A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix} and\ I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{vmatrix} -5-\lambda & -3 \\ 2 & 0-\lambda \end{vmatrix} = 0 \quad (-5-\lambda)(-\lambda)+6=0 \Rightarrow \lambda^2+5\lambda+6=0$

$So, A^2+5A+6I=0$

By Caley Hamilton theorem

$\Rightarrow A^2 = -5A-6I$

Multiplying by A on both sides, we have

$A^3 = -5A^2 - 6A$

$\Rightarrow \quad A^3 = -5(-5A-6I)-6A$

$= 19A + 30I$

22. b)

The augumented matrix for the system or equations is

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 4 & 6 & : & 20 \\ 1 & 4 & \lambda & : & \mu \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 4 & 6 & : & 20 \\ 0 & 0 & \lambda-6 & : & \mu-20 \end{bmatrix} [R_3 \to R_3 - R_2]$$

If λ = 6 and $\mu \neq 20$ then

$\rho(A:B) = 3 and\ \rho(A) = 2$

$\because \quad \rho(A:B) \neq \rho(A)$

$\therefore$ Given system of equations has no solution

For $\lambda = 6 and\ \mu \neq 20$

23. c)

A skew – symmetric matrix is a square matrix whose transpose is equal to its negative value i.e. A^T = -A the eigen values of a skew- symmetric matrix always come in pairs $\pm\lambda$ (except in odd-dimensional case where there is an additional unpaired 0 eigen value). For a

Career Avenues

real skew symmetric matrix, the non-zero eigen values are all pure imaginary and thus are of the form $i\lambda,.-i\lambda,.i\lambda_2,-i\lambda_2,....$ where each of the λ_K is real.

24. d)
Sum of the eigen values are the sum of the diagonal element of the matrix
Sum of the diagonal element= 3-1-1=1
In option d) sum of eigen value =3-1+3j-1-3j= 3-1-1=1
Also, since matrix is real, eigen values are either real or complex conjugate.

25. b) For the given system of equations
the rank of coefficient matrix [A]=1 and rank of augmented matrix is [A: B] = 2, So
No solution exists.

26. c) The Characteristics equation of A will be |P-SI|=0

$$[sI-P]=\begin{bmatrix} s-P_{11} & -P_{12} \\ -P_{21} & s-P_{22} \end{bmatrix}$$

$$[sI-P]=0$$

$$(s-P_{11})(s-P_{22})-P_{12}P_{21}=0$$

$$s^2-(P_{11}+P_{22})s+P_{11}P_{22}-P_{12}P_{21}=0$$

Since, one of the eigenvalues is zero, therefore
Putting s = 0

$$\boxed{P_{11}P_{22}-P_{12}P_{21}=0}$$

Which is the desired condition

27. c)

$$M=\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \quad M-\lambda I=\begin{bmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix}$$

Given eigen vector

$$\therefore \begin{bmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix}\begin{bmatrix} 101 \\ 101 \end{bmatrix}=0$$

$$\Rightarrow \quad (4-\lambda)(101)+2(101)=0$$

$$\Rightarrow \quad 4-\lambda+2=0$$

$$\lambda=6$$

Career Avenues